



AOS-CX 10.17.xxxx ACLs and Classifier Policies Guide (For 4100i, 6000, 6100 Switch series)

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About this document

This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing HPE Aruba Networking switches on a network.

Subtopics

Applicable products

Latest version available online

Command syntax notation conventions

About the examples

Identifying switch ports and interfaces

Applicable products

This document applies to the following products:

- HPE Aruba Networking 4100i Switch Series (JL817A, JL818A)
- HPE Aruba Networking 6000 Switch Series (R8N85A, R8N86A, R8N87A, R8N88A, R8N89A, R9Y03A, S4R20A, S4R21A, S4R22A, S4R23A, S4R24A, S4R25A, S4R26A, S4R27A, S4R28, S4R29A)
- HPE Aruba Networking 6100 Switch Series (JL675A, JL676A, JL677A, JL678A, JL679A)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
example - text	Identifies commands and their options and operands , code examples, filenames, pathnames, and output displayed in a command window. Items that appear li

Convention	Usage
	ke the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example - text	In code and screen examples, indicates text entered by a user.
Any of the following: • <example - text> • <example - text> • example - text • example - text	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: • For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. • For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: • In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. • In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term **switch**, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME) #
```

Indicates the configuration context for a feature. For example:

```
switch(config-if) #
```

Identifies the **interface** context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100) #
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>) #
```

Where <VLAN-ID> is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format: **member/slot/port**.

On the HPE Aruba Networking 4100i Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

On the HPE Aruba Networking 6000 and 6100 Switch Series

- member: Always 1. VSF is not supported on this switch.
- slot: Always 1. This is not a modular switch, so there are no slots.
- port: Physical number of a port on the switch.

For example, the logical interface **1/1/4** in software is associated with physical port 4 on the switch.

Access Control Lists

Access Control Lists (ACLs) let a network administrator permit or deny passage of traffic based on network addresses, protocols, service ports, and other packet attributes. ACLs are composed of one or more Access Control Entries (called ACEs). Each ACE defines a filter criteria and an action, either **permit** or **deny**. If the traffic matches the filter criteria, the specified action is taken. The **permit** action permits the traffic to continue through the switch. The **deny** action causes the traffic to be discarded (dropped). ACEs can also log or count matching traffic.

Three ACL types are supported; IPv4, IPv6, and MAC. Each ACL type is focused on relevant frame or packet characteristics.

ACLs must be applied (using an **apply access-list** command) to take effect. ACLs can be applied to interfaces (including LAGs), VLANs, or the Control Plane.

Access Control Entries (ACEs) are listed according to priority by sequence number and processed in lowest to highest sequence number order. Each ACE attempts to match on one or more attributes of the particular traffic type. Attempted ACE matching ceases upon the first successful match. For a match to be considered successful, a packet must match all the criteria, qualifiers, and attributes of a particular ACE. Higher-numbered ACEs are only processed if no lower-numbered ACE matches. If the traffic matches no ACE in the entire ACL, the default action **deny** is taken, causing the traffic to be discarded (dropped).

When defining an ACE, if the sequence number is omitted, the ACE is auto-assigned a new sequence number that is 10 greater than the existing highest ACE sequence number. The first auto-assigned sequence number is 10. If you choose to include the ACE sequence numbers, you can use any number you like, however it is suggested that you follow the practice of entering them as 10, 20, 30, and so on. Regardless of the order in which ACEs are entered, they are stored in low-to-high sequence number order. If you enter three ACEs numbered 10, 30, 20, when creating an ACL, the ACEs are stored in the ACL as 10, 20, 30.

This simple ACL definition permits traffic passage for a particular address range and otherwise counts all nonmatching (dropped) traffic:

```
switch(config) # access-list ip network-A-udp-only  
switch(config-acl-ip) # 10 permit udp any 172.16.1.0/24  
switch(config-acl-ip) # 20 deny any any any count
```

```
switch(config-acl-ip) # exit
```

The main traffic characteristics that ACEs can filter on are as follows (see the full list in the ACE parameters list of the ACL commands):

- Protocol such as: ICMP, TCP, UDP
- Source and/or destination addresses (IPv4, IPv6, or MAC)
- Source and/or destination TCP/UDP ports (if applicable to the specified protocol)

A few real-world uses of ACLs are as follows:

- Restrict traffic arriving on a port, destined to a particular address or subnet by applying an ACL that matches on a destination IP address or an IP address and a mask.
- Prevent certain protocols from using a particular multicast MAC address (advertising through a port) by applying an ACL that matches on the destination MAC address.
- Prevent any IP host from accessing a particular IP port/application on a specific server by applying an ACL that matches on IP addresses and Layer 4 port.



NOTE

See also [ACL and Policy Hardware Resource Considerations](#).

Subtopics

ACL usage tips

ACL and ACE-related tasks

Active ACL configuration versus user-specified configuration

ACL commands

ACL usage tips

When using the `access-list ip` or `access-list ipv6` commands, if you enter an existing `ACL-NAME`, the existing ACL is modified as follows:

- Any ACE entered with a new sequence-number creates an additional ACE.
- Any ACE entered with an existing sequence-number replaces the existing ACE.

If you modify an ACL that has already been applied, it is possible that packets, blocked by the previous ACL, will briefly pass through the switch during the ACL reconfiguration.



NOTE

In a highly secure environment, it is safest to first bring down interfaces and VLANs to which an ACL has been applied before modifying the ACL. Then bring the targets of ACL application back up after completing the ACL modification. Respecting this recommendation ensures that an ACL is never partially programmed while traffic is passing through the switch.

About applying ACLs to interfaces or LAGs

You can apply an ACL to an interface or LAG to affect or control the traffic arriving on that interface or LAG. A given interface or LAG supports the application of a single ACL per type. ACLs can be applied to interfaces or LAGs as follows:

- One MAC ACL inbound
- One IPv4 ACL inbound
- One IPv6 ACL inbound

If you apply an ACL of a particular type that is already in use, the switch replaces the current ACL with the new ACL.

Sequence numbering

If no sequence number is specified, the software appends new ACEs to the end of the ACL with a sequence number equal to the highest ACE currently in the list plus 10.

The sequence numbers may be resequenced using the `access-list resequence` command.

ACL Logging

On the 4100, 6000, and 6100 Switch Series: ACL Logging will not function properly on a VSF stack for packets received on a member that is not the conductor. The ACL name, sequence number, and action for the matching ACE will be not be logged.

Deny ACLs

If multiple ACLs of different types are applied in the same direction, a deny ACE, whether explicit or implicit, in one ACL overrides a permit ACL in another. A deny ACE is an ACE within an ACL that uses the `deny` action keyword.

Denied ping requests

A ping request is denied when an ACL is applied on ingress unless the request is explicitly permitted.

```
switch# ping 100.1.2.10
PING 100.1.2.10 (100.1.2.10) 100(128) bytes of data.
ping: sendmsg: Operation not permitted
```

```
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
ping: sendmsg: Operation not permitted
```

ACL failure behaviors

- In the event of a failed ACL application to a VLAN or VLAN interface during a switch reboot, all of the ports will shut down. The switch must be restarted to recover from the failure. Modifying the VLAN, VLAN interface, or ACL configuration will not cause the switch to be restarted.

ACLs and VXLAN Traffic

ACL and ACE-related tasks

Common ACL and ACE-related tasks are as follows:

Task	Command name	Example
Creating an IPv4 ACL	<code>access-list ip</code>	<pre>access-list ip MY_IP_ACL 10 permit udp any 172.16.1.0/24 20 permit tcp 172.16.2.0/16 gt 1023 any 30 deny any any any count</pre>
Creating an IPv6 ACL	<code>access-list ipv6</code>	<pre>access-list ipv6 MY_IPV6_ACL 10 permit udp any 2001::1/64 20 permit tcp 2001:2011::1/64 any 30 deny any any any count</pre>
Creating a MAC ACL	<code>access-list mac</code>	<pre>access-list mac MY_MAC_ACL 10 permit any any appletalk vlan 40</pre>

Task	Command name	Example
		20 deny any any any count
Applying an IPv6 ACL to an interface	apply access-list (to interface or LAG)	interface 1/1/1 apply access-list ipv6 MY_IPV6_ACL in
Applying an IPv4 ACL to a LAG	apply access-list (to interface or LAG)	interface lag 100 apply access-list ip MY_IP_ACL in
Applying an IPv4 ACL to a VLAN	apply access-list (to VLAN)	vlan 10 apply access-list ip MY_IP_ACL in
Applying a MAC ACL to a VLAN	apply access-list (to VLAN)	vlan 40 apply access-list mac MY_MAC_ACL in
Removing application of an ACL from an interface	apply access-list (to interface or LAG)	interface 1/1/1 no apply access- list ipv6 MY_IPV6_ACL in
Removing application of an ACL from a VLAN	apply access-list (to interface VLAN)	vlan 40 no apply access- list mac MY_MAC_ACL in
Showing all ACLs	show access-list	show access-list
Showing all IPv6 ACLs	show access-list	show access-list ipv6
Showing all ACLs applied to interface 1/1/1	show access-list	show access-list interface 1/1/1
Showing all ACLs applied to VLAN 10	show access-list	show access-list vlan 10
Showing all ACLs applied to the Control Plane	show access-list control-plane	show access-list control-plane
Showing a particular ACL	show access-list	show access-list ip MY_ACL
Showing an ACL as commands	show access-list	show access-list ip MY_ACL commands

Task	Command name	Example
Showing ACL hit counts for an ACL applied to an interface	<code>show access-list hitcounts</code>	<code>show access-list hitcounts ip MY_ACL interface 1/1/1</code>
Showing ACL hit counts for an ACL applied to a VLAN	<code>show access - list hitcounts</code>	<code>show access-list hitcounts ip MY_ACL vlan 10</code>
Clearing ACL hit counts	<code>clear access-list hitcounts</code>	<code>clear access-list hitcounts ip MY_ACL vlan 10</code>
Copying an ACL	<code>access-list copy</code>	<code>access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2</code>
Resequencing the ACEs of an ACL	<code>access-list resequence</code>	<code>access-list ip MY_IP_ACL resequence 1 1</code>
Resetting an ACL	<code>access-list reset</code>	<code>access-list ip MY_IP_ACL reset</code>
Setting the ACL log timer frequency	<code>access-list log-timer</code>	<code>access-list log-timer 30</code>

Active ACL configuration versus user-specified configuration

The **show access-list** command shows the active configuration of the switch. The active configuration is the ACLs that have been configured and accepted by the system. The active configurations are the interfaces on which the ACLs have successfully been programmed in the hardware.

The output of the `show access-list` command with the **configuration** parameter shows the ACLs that have been configured. The output of this command may not be the same as what was programmed in the hardware or what is active on the switch. The situation might occur because of one or more of the following:

- Unsupported command parameters might have been configured.
- Unsupported applications might have been specified.
- Applying an ACL might have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run the `show access-list` command with the `configuration` parameter.

If the active ACLs and configured ACLs are not the same, the switch shows a warning message in the output of the show command:

```
! access-list ip MY_IP_ACL user configuration does not match active
configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
```

If the configured ACL is processing, the switch shows an in-progress warning.

```
! access-list ip MY_IP_ACL user configuration currently being processed
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
```

If the switch shows a warning message or in-progress message, additional changes can be made until the error message is no longer shown in the show command, or you can run the `access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset` command. The `access-list reset` command changes the user-specified configuration to match the active configuration. For details, see [access-list reset](#).



NOTE

The `show running-config` command also shows a warning about ACLs that are in progress or failed.

Examples

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Showing the user-specified configuration:

```
switch(config)# do show access-list ip TEST_ACL
  10 permit tcp 172.16.2.0/16 any ack
  interface 1/1/1
  ! access-list ip TEST_ACL user configuration does not match active
configuration.
  ! run 'show access-list [commands]' to display active access-list
configuration.
  apply access-list ip TEST_ACL in
switch(config)# do show access-list commands
access-list ip TEST_ACL
  10 permit tcp 172.16.2.0/16 any ack
! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list all reset' to reset all access-lists to match active
configuration.
switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
  10 permit tcp 172.16.2.0/16 any ack
```

```

! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list all reset' to reset all access-lists to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in
switch(config)# do show access-list
Type          Name
Sequence Comment
              Action                      L3 Protocol
              Source IP Address          Source L4 Port(s)
              Destination IP Address      Destination L4 Port(s)
              Additional Parameters
-----
---
IPv4          TEST_ACL
              10 permit                      tcp
              172.16.2.0/16
              any
              ack

```

Resetting the user-specified configuration to match the active configuration:

```
switch(config)# access-list all reset
```

Showing the updated user-specified configuration:

```
switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/16 any ack

```

ACL commands

Select a command from the list in the left navigation menu.



NOTE

On the HPE Aruba Networking 6000 and 6100 Switch Series, only the vrf named **default** is available. Replace any references to the **mgmt** or other VRFs with **default**.



NOTE

On the HPE Aruba Networking 4100i Switch Series, only the vrf named **default** is available. Replace any references to the **mgmt** or other VRFs with **default**.

Subtopics

[access-list copy](#)

[access-list ip](#)

[access-list ipv6](#)

[access-list log-timer](#)

[access-list mac](#)

[access-list resequence](#)

[access-list reset](#)

[access-list secure-update](#)

[ACL application](#)

[apply access-list control-plane](#)

[apply access-list \(to interface or LAG\)](#)

[apply access-list \(to VLAN\)](#)

[clear access-list hitcounts](#)

[clear access-list hitcounts control-plane](#)

[show access-list](#)

[show access-list control-plane](#)

[show access-list hitcounts](#)

[show access-list hitcounts control-plane](#)

[show access-list secure-update](#)

[show capacities](#)

[show capacities-status](#)

access-list copy

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> copy <DESTINATION-ACL>
```

Description

Copies an IPv4, IPv6, or MAC ACL to a new destination ACL or overwrites an existing ACL.

Parameter	Description
<code>{ip ipv6 mac}</code>	Specifies the type of ACL.
<code><ACL-NAME></code>	Specifies the name of the ACL to be copied.
<code><DESTINATION-ACL></code>	Specifies the name of the destination ACL.

Examples

Copying MY_IP_ACL to MY_IP_ACL2:

```
switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config-acl-ip)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv4	MY_IP_ACL									
		1	permit		udp	any		172.16.1.0/255.255.255.0		
		2	permit		tcp	172.16.2.0/255.255.0.0	> 1023	any		
		3	permit		tcp	172.26.1.0/255.255.255.0		any		
				dscp: AF11						
				ack						
				syn						

```

4 deny                                any
  any
  any
  Hit-counts: enabled
-----
---
IPv4      MY_IP_ACL2
1 permit                                udp
  any
  172.16.1.0/255.255.255.0
2 permit                                tcp
  172.16.2.0/255.255.0.0          > 1023
  any
3 permit                                tcp
  172.26.1.0/255.255.255.0
  any
  dscp: AF11
  ack
  syn
4 deny                                any
  any
  any
  Hit-counts: enabled

```

Copying MY_IPV6_ACL to MY_IPV6_ACL2:

```

switch(config)# access-list ipv6 MY_IPV6_ACL copy MY_IPV6_ACL2
switch(config-acl-ip)# exit
switch(config)# do show access-list
Type      Name
Sequence  Comment
          Action                                L3 Protocol
          Source IP Address                    Source L4 Port(s)
          Destination IP Address                Destination L4 Port(s)
          Additional Parameters
-----
---
IPv6      MY_IPV6_ACL
1 permit                                udp
  any
  2001::1/64
2 Permit all TCP ephemeral ports
  permit                                tcp
  2001:2001::2:1          > 1023
  any

```

```

3 permit                                tcp
  2001:2011::1/64
  any
4 deny                                any
  any
  any
Hit-counts: enabled
-----
---
IPv6      MY_IPV6_ACL2
1 permit                                udp
  any
  2001::1/64
2 Permit all TCP ephemeral ports
  permit                                tcp
  2001:2001::2:1                        > 1023
  any
3 permit                                tcp
  2001:2011::1/64
  any
4 deny                                any
  any
  any
Hit-counts: enabled

```

Copying MY_MAC_ACL to MY_MAC_ACL2:

```

switch(config)# access-list mac MY_MAC_ACL copy MY_MAC_ACL2
switch(config-acl-mac)# exit
switch(config)# do show access-list
Type      Name
Sequence Comment
          Action
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
---
MAC      MY_MAC_ACL
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333

```

```

QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

```

MAC      MY_MAC_ACL2
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

Type	Name
Sequence	Comment
	Action
	EtherType
	Source MAC Address
	Destination MAC Address
	Additional Parameters

```

MAC      MY_MAC_ACL
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc

```

```

1111.2222.3333
QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

```

MAC      MY_MAC_ACL2
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-1 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

Command History

Release	Modification
---------	--------------

10.07 or earlier	- -
------------------	-----

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

access-list ip

Syntax

Syntax to create an IPv4 ACL and enter its context. Plus syntax to remove an ACL:

```
access-list ip <ACL-NAME>
no access-list ip <ACL-NAME>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols **ah**, **eigrp**, **esp**, **gre**, **igmp**, **l2tp**, **ospf**, **pim**, **vrrp**, (**ip** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|ip|ah|gre|esp|igmp|ospf|pim|
    <IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocols **sctp**, **tcp**, **udp**:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>} ] }
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
```

```
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[ tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for creating or removing ACEs for protocol `icmp`:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[ tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>
```

Syntax (within the ACL context) for ACE comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment
```

Description

Creates an IPv4 Access Control List (ACL) comprised of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Parameter	Description
<code><ACL-NAME></code>	Specifies the name of this ACL.
<code><SEQUENCE-NUMBER></code>	Specifies a sequence number for the ACE. Range: 1 to 4294967295.
<code>{permit deny}</code>	Specifies whether to permit or deny traffic matching this ACE.
	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.

Parameter	Description
<code><IP-PROTOCOL-NUM></code>	
<pre>{any <SRC-IP-ADDRESS>[/ <PREFIX-LENGTH> <SUBNET-MASK>]} }</pre>	Specifies the source IPv4 address. • any - specifies any source IPv4 address. • <SRC - IP - ADDRESS> - specifies the source IPv4 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<pre>{any <DST-IP-ADDRESS>[/ <PREFIX-LENGTH> <SUBNET-MASK>]} }</pre>	Specifies the destination IPv4 address. • any - specifies any destination IPv4 address. • <DST - IP - ADDRESS> - specifies the destination IPv4 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<pre>[{eq gt lt} <PORT> range <MIN-PORT> <MAX-PORT>]</pre>	Specifies the port, or port range. Port numbers are in the range of 0 to 65535. • eq <PORT> - specifies the Layer 4 port. • gt <PORT> - specifies any Layer 4 port greater than the indicated port. • lt <PORT> - specifies any Layer 4 port less than the indicated port. • range <MIN - PORT> <MAX - PORT> - specifies the Layer 4 port range.  NOTE Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.
<code>urg</code>	Specifies matching on the TCP Flag: Urgent .
<code>ack</code>	Specifies matching on the TCP Flag: Acknowledgment .

Parameter	Description
psh	Specifies matching on the TCP Flag: Push buffered data to receiving application.
rst	Specifies matching on the TCP Flag: Reset the connection.
syn	Specifies matching on the TCP Flag: Synchronize sequence numbers.
fin	Specifies matching on the TCP Flag: Finish connection.
established	Specifies matching on the TCP Flag: Established connection.
[icmp-type {echo echo-reply <ICMP-TYPE-VALUE>}]	Specifies the ICMP type. • echo - specifies an ICMP echo request packet. • echo - reply - specifies an ICMP echo reply packet. • <ICMP - TYPE - VALUE> - specifies an ICMP type value. Range: 0 to 255.
[icmp-code <ICMP-CODE-VALUE>]	Specifies the ICMP code value. Range: 0 to 255.
dscp DSCP-SPECIFIER>	Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP - VALUE> (0 to 63) or one of these keywords : • AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability) • AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability) • AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability) • AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability) • AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability) • AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability) • AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability) • AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability) • AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)

Parameter	Description
	• AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability) • AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability) • CS0 - DSCP 0 (Class Selector 0: Default) • CS1 - DSCP 8 (Class Selector 1: Scavenger) • CS2 - DSCP 16 (Class Selector 2: OAM) • CS3 - DSCP 24 (Class Selector 3: Signaling) • CS4 - DSCP 32 (Class Selector 4: Real time) • CS5 - DSCP 40 (Class Selector 5: Broadcast video) • CS6 - DSCP 48 (Class Selector 6: Network control) • CS7 - DSCP 56 (Class Selector 7) • EF - DSCP 46 (Expedited Forwarding)
<code>ip-precedence <IP-PRECEDENCE-VALUE></code>	Specifies an IP precedence value. Range: 0 to 7.
<code>tos <TOS-VALUE></code>	Specifies the Type of Service value. Range: 0 to 31.
<code>fragment</code>	Specifies a fragment packet.
<code>vlan <VLAN-ID></code>	Specifies VLAN tag to match on. 802.1Q VLAN ID.
	 NOTE This parameter cannot be used in any ACL that will be applied to a VLAN.
<code>count</code>	Keeps the hit counts of the number of packets matching this ACE.
<code>log</code>	Keeps a log of the number of packets matching this ACE. Works with both <code>permit</code> and <code>deny</code> actions. Works with ACLs applied on ingress, or Control Plane, but not with ACLs applied on egress.
<code>[<SEQUENCE-NUMBER>] comment <TEXT-STRING></code>	Adds a comment to an ACE. The no form removes only the comment from the ACE.

Usage

- If the <IP-PROTOCOL-NUM> parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with **log** option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log for all the ACLs that were matched, regardless of type.

Examples

Creating an IPv4 ACL with four entries:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 10 permit udp any 172.16.1.0/24
switch(config-acl-ip)# 20 permit tcp 172.16.2.0/16 gt 1023 any
switch(config-acl-ip)# 30 permit tcp 172.26.1.0/24 any syn ack dscp 10
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit
switch(config)# show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------

IPv4	MY_IP_ACL									
		10	permit		udp	any				
						172.16.1.0/255.255.255.0				
		20	permit		tcp	172.16.2.0/255.255.0.0	> 1023			
						any				
		30	permit		tcp	172.26.1.0/255.255.255.0				
						any				
						dscp: AF11				
						ack				
						syn				
		40	deny		any					
						any				

```
any
Hit-counts: enabled
```

Adding a comment to an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ip)# exit
switch(config)# show access-list
```

Type	Name		
Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		

```
-----
---
IPv4      MY_IP_ACL
10 permit                                udp
    any
    172.16.1.0/255.255.255.0
20 Permit all TCP ephemeral ports
    permit                                tcp
    172.16.2.0/255.255.0.0                > 1023
    any
30 permit                                tcp
    172.26.1.0/255.255.255.0
    any
    dscp: AF11
    ack
    syn
40 deny                                    any
    any
    any
Hit-counts: enabled
```

Removing a comment from an existing IPv4 ACE:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 20 comment
switch(config-acl-ip)# exit
switch(config)# show access-list
```

Type	Name		
Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)

Additional Parameters

```

-----
---
IPv4      MY_IP_ACL
10 permit                                udp
    any
    172.16.1.0/255.255.255.0
20 permit                                tcp
    172.16.2.0/255.255.0.0          > 1023
    any
30 permit                                tcp
    172.26.1.0/255.255.255.0
    any
    dscp: AF11
    ack
    syn
40 deny                                    any
    any
    any
Hit-counts: enabled

```

Adding an ACE (insert line 25) to an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.16.2.0/16 any
switch(config-acl-ip)# exit
switch(config)# show access-list
Type      Name
Sequence Comment
Action                                L3 Protocol
Source IP Address                    Source L4 Port(s)
Destination IP Address                Destination L4 Port(s)
Additional Parameters
-----
---
IPv4      MY_IP_ACL
10 permit                                udp
    any
    172.16.1.0/255.255.255.0
20 permit                                tcp
    172.16.2.0/255.255.0.0          > 1023
    any
25 permit                                icmp
    172.16.2.0/255.255.0.0 any
30 permit                                tcp

```

```

172.26.1.0/255.255.255.0
any
dscp: AF11
ack
syn
40 deny                                any
any
any
Hit-counts: enabled

```

Replacing an ACE in an existing IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# 25 permit icmp 172.17.1.0/16 any
switch(config-acl-ip)# exit
switch(config)# show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv4	MY_IP_ACL	10	permit	udp		any		172.16.1.0/255.255.255.0		
		20	permit	tcp		172.16.2.0/255.255.0.0	> 1023	any		
		25	permit	icmp		172.17.1.0/255.255.0.0				
		30	permit	tcp		172.26.1.0/255.255.255.0		any		
										dscp: AF11
										ack
										syn
		40	deny	any		any		any		
										Hit-counts: enabled

```

Type      Name
Sequence  Comment

```

	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	MY_IP_ACL	
10	permit	udp
	any	
	172.16.1.0/255.255.255.0	
20	permit	tcp
	172.16.2.0/255.255.0.0	> 1023
	any	
25	permit	icmp
	172.17.1.0/255.255.0.0	
30	permit	tcp
	172.26.1.0/255.255.255.0	
	any	
	dscp: AF11	
	ack	
	syn	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

Removing an ACE from an IPv4 ACL:

```
switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# no 25
switch(config-acl-ip)# exit
switch(config)# show access-list
```

Type	Name	
	Sequence	Comment
	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	MY_IP_ACL	
10	permit	udp
	any	
	172.16.1.0/255.255.255.0	
20	permit	tcp

```

172.16.2.0/255.255.0.0          > 1023
any
30 permit                      tcp
172.26.1.0/255.255.255.0
any
dscp: AF11
ack
syn
40 deny                        any
any
any
Hit-counts: enabled
Type      Name
Sequence  Comment
Action    L3 Protocol
Source IP Address    Source L4 Port(s)
Destination IP Address    Destination L4 Port(s)
Additional Parameters
-----
---
IPv4      MY_IP_ACL
10 permit                      udp
any
172.16.1.0/255.255.255.0
20 permit                      tcp
172.16.2.0/255.255.0.0        > 1023
any
30 permit                      tcp
172.26.1.0/255.255.255.0
any
dscp: AF11
ack
syn
40 deny                        any
any
any
Hit-counts: enabled

```

Copy an IPv4 ACL:

```

switch(config)# access-list ip MY_IP_ACL copy MY_IP_ACL2
switch(config)# show access-list
Type      Name
Sequence  Comment
Action    L3 Protocol

```

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	MY_IP_ACL	
10	permit	udp
	any	
	172.16.1.0/255.255.255.0	
20	permit	tcp
	172.16.2.0/255.255.0.0	> 1023
	any	
30	permit	tcp
	172.26.1.0/255.255.255.0	
	any	
	dscp: AF11	
	ack	
	syn	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

IPv4	MY_IP_ACL2	
10	permit	udp
	any	
	172.16.1.0/255.255.255.0	
20	permit	tcp
	172.16.2.0/255.255.0.0	> 1023
	any	
30	permit	tcp
	172.26.1.0/255.255.255.0	
	any	
	dscp: AF11	
	ack	

```

        syn
40
    deny                                any
    any
    any
    Hit-counts: enabled switch(config)# access-list ip MY_IP_ACL
copy MY_IP_ACL2
switch(config)# show access-list
Type          Name
Sequence Comment
Action                               L3 Protocol
Source IP Address                     Source L4 Port(s)
Destination IP Address                 Destination L4 Port(s)
Additional Parameters

-----
---
IPv4          MY_IP_ACL
10
    permit                                udp
    any
    172.16.1.0/255.255.255.0
20
    permit                                tcp
    172.16.2.0/255.255.0.0                > 1023
    any
30
    permit                                tcp
    172.26.1.0/255.255.255.0
    any
    dscp: AF11
    ack
    syn
40
    deny                                any
    any
    any
    Hit-counts: enabled

-----
---
IPv4          MY_IP_ACL2
10
    permit                                udp
    any

```

```

172.16.1.0/255.255.255.0
20
  permit                                tcp
  172.16.2.0/255.255.0.0                > 1023
  any
30
  permit                                tcp
  172.26.1.0/255.255.255.0
  any
  dscp: AF11
  ack
  syn
40
  deny                                any
  any
  any
Hit-counts: enabled

```

Removing an IPv4 ACL:

```
switch(config)# no access-list ip MY_IP_ACL
```

```
switch(config)# show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv4	MY_IP_ACL2	1	permit	any	udp	172.16.1.0/255.255.255.0				
		2	permit	172.16.2.0/255.255.0.0	tcp		> 1023			
		3	permit	172.26.1.0/255.255.255.0	tcp					
				dscp: AF11						
				ack						
				syn						
		4	deny	any	any					
				any						

```

any
Hit-counts: enabled
switch(config-acl-ip)# permit sctp any group my_port_group any ip-option ?
any Any IP option

```

Command History

Release	Modification
10.17.1000	Added support for EIGRP, L2TP and VRRP protocols.
10.12	Allow ACLs applied to the Control Plane to be logged.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config The access - list ip < ACL - NAME > command takes you into the named ACL context where you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list ipv6

Syntax

Syntax to create an IPv6 ACL and enter its context. Plus syntax to remove an ACL:

```

access-list ipv6 <ACL-NAME>
no access-list ipv6 <ACL-NAME>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols **ah**, **eigrp**, **esp**, **gre**, **l2tp**, **ospf**, **pim**, **vrrp** (**ipv6** is available as an alias for **any**):

```

[ <SEQUENCE-NUMBER>]
{permit|deny}
{any|ah|eigrp|esp|gre|ipv6|l2tp|ospf|pim|vrrp| <IP-PROTOCOL-NUM>}

```

```

{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for creating or removing ACEs for protocols **sctp**, **tcp**, **udp**:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]]}
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for creating or removing ACEs for protocol **icmpv6**:

```

[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-
VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>

```

Syntax (within the ACL context) for ACE comments:

```

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment

```

Description

Creates an IPv6 Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence number. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Parameter	Description
<code><ACL-NAME></code>	Specifies the name of this ACL.
<code><SEQUENCE-NUMBER></code>	Specifies a sequence number for the ACE. Range: 1 to 4294967295.
<code>{permit deny}</code>	Specifies whether to permit or deny traffic matching this ACE.
<code><IP-PROTOCOL-NUM></code>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
<code>{any <SRC-IP-ADDRESS>[/<PREFIX-LENGTH>]}</code>	Specifies the source IPv6 address. • any - specifies any source IPv6 address. • <code><SRC - IP - ADDRESS></code> - specifies the source IPv6 host address. ◦ <code><PREFIX - LENGTH></code> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.
<code>{any <DST-IP-ADDRESS>[/<PREFIX-LENGTH>]}</code>	Specifies the destination IPv6 address. • any - specifies any destination IPv6 address. • <code><DST - IP - ADDRESS></code> - specifies the destination IPv6 host address. ◦ <code><PREFIX - LENGTH></code> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 128.
<code>[{eq gt lt} <PORT> range <MIN-PORT> <MAX-PORT>]</code>	Specifies the port, or port range. Port numbers are in the range of 0 to 65535. • eq <code><PORT></code> - specifies the Layer 4 port. • gt <code><PORT></code> - specifies any Layer 4 port greater than the indicated port. • lt <code><PORT></code> - specifies any Layer 4 port less than the indicated port.

Parameter	Description
	range <MIN - PORT> <MAX - PORT> - specifies the Layer 4 port range.  NOTE Upon application of the ACL, ACEs with L4 port ranges may consume more than one hardware entry.
<code>[icmp-type {echo echo-reply <ICMP-TYPE-VALUE>}]</code>	Specifies the ICMP type. echo - specifies an ICMP echo request packet. echo-reply - specifies an ICMP echo reply packet. <ICMP - TYPE - VALUE> - specifies an ICMP type value. Range: 0 to 255.
<code>[icmp-code <ICMP-CODE-VALUE>]</code>	Specifies the ICMP code value. Range: 0 to 255.
<code>dscp DSCP-SPECIFIER></code>	Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP - VALUE> (0 to 63) or one of these keywords: AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability) AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability) AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability) AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability) AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability) AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability) AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability) AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability) AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability) AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)

Parameter	Description
	• AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability) • CS0 - DSCP 0 (Class Selector 0: Default) • CS1 - DSCP 8 (Class Selector 1: Scavenger) • CS2 - DSCP 16 (Class Selector 2: OAM) • CS3 - DSCP 24 (Class Selector 3: Signaling) • CS4 - DSCP 32 (Class Selector 4: Real time) • CS5 - DSCP 40 (Class Selector 5: Broadcast video) • CS6 - DSCP 48 (Class Selector 6: Network control) • CS7 - DSCP 56 (Class Selector 7) • EF - DSCP 46 (Expedited Forwarding)
<code>ip-precedence <IP-PRECEDENCE-VALUE></code>	Specifies an IP precedence value. Range: 0 - 7.
<code>tos <TOS-VALUE></code>	Specifies the Type of Service value. Range: 0 - 31.
<code>vlan <VLAN-ID></code>	Specifies VLAN tag to match on. 802.1Q VLAN ID.
	 NOTE This parameter cannot be used in any ACL that will be applied to a VLAN.
<code>count</code>	Keeps the hit counts of the number of packets matching this ACE.
<code>log</code>	Keeps a log of the number of packets matching this ACE. Works with both <code>permit</code> and <code>deny</code> actions. Works with ACLs applied on ingress, or Control Plane, but not with ACLs applied on egress.
<code>[<SEQUENCE-NUMBER>] comment <TEXT-STRING></code>	Adds a comment to an ACE. The no form removes only the comment from the ACE.

Usage

- If the `<IP-PROTOCOL-NUM>` parameter is used instead of a protocol name, ensure that any needed ACE-definition parameters specific to the selected protocol are also provided.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets

matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating an IPv6 ACL with four entries:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 10 permit udp any 2001::1/64
switch(config-acl-ipv6)# 20 permit tcp 2001:2001::2:1/128 gt 1023 any
switch(config-acl-ipv6)# 30 permit tcp 2001:2011::1/64 any
switch(config-acl-ipv6)# 40 deny any any any count
switch(config-acl-ipv6)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL	10	permit	udp		any		2001::1/64		
		20	permit	tcp		2001:2001::2:1	> 1023	any		
		30	permit	tcp		2001:2011::1/64		any		
		40	deny	any		any		any		
										Hit-counts: enabled

Adding a comment to an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 20 comment Permit all TCP ephemeral ports
switch(config-acl-ipv6)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol
		20	comment Permit all TCP ephemeral ports		

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
10	permit	udp
	any	
	2001::1/64	
20	Permit all TCP ephemeral ports	
	permit	tcp
	2001:2001::2:1	> 1023
	any	
30	permit	tcp
	2001:2011::1/64	
	any	
40	deny	any
	any	
	any	
	Hit-counts: enabled	

Removing a comment from an existing IPv6 ACE:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 20 comment
switch(config-acl-ipv6)# exit
switch(config)# do show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv6	MY_IPV6_ACL		
10	permit	udp	
	any		
	2001::1/64		
20	permit	tcp	
	2001:2001::2:1	> 1023	
	any		
30	permit	tcp	
	2001:2011::1/64		
	any		

```

40 deny
    any
    any
Hit-counts: enabled

```

Adding an ACE to an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::1/64 any
switch(config-acl-ipv6)# exit
switch(config)# do show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	MY_IPV6_ACL	10	permit	any	udp	2001::1/64				
		20	permit	2001:2001::2:1	tcp		> 1023			
		25	permit	2001::1/64	icmpv6					
		30	permit	2001:2011::1/64	tcp					
		40	deny	any	any					
										Hit-counts: enabled

Replacing an ACE in an existing IPv6 ACL:

```

switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# 25 permit icmpv6 2001::2:1/64 any
switch(config-acl-ipv6)# exit
switch(config)# do show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)
		25	permit	2001::2:1/64	icmpv6		

	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
10	permit	udp
	any	
	2001::1/64	
20	permit	tcp
	2001:2001::2:1	> 1023
	any	
25	permit	icmpv6
	2001::2:1/64	
	any	
30	permit	tcp
	2001:2011::1/64	
	any	
40	deny	any
	any	
	any	
	Hit-counts: enabled	
Type	Name	
Sequence	Comment	
	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv6	MY_IPV6_ACL	
10	permit	udp
	any	
	2001::1/64	
20	permit	tcp
	2001:2001::2:1	> 1023
	any	
25	permit	icmpv6
	2001::2:1/64	
	any	
30	permit	tcp
	2001:2011::1/64	
	any	
40	deny	any

```
any
any
Hit-counts: enabled
```

Removing an ACE from an IPv6 ACL:

```
switch(config)# access-list ipv6 MY_IPV6_ACL
switch(config-acl-ipv6)# no 25
switch(config-acl-ipv6)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------

```
---
IPv6      MY_IPV6_ACL
10 permit                                udp
    any
    2001::1/64
20 permit                                tcp
    2001:2001::2:1                       > 1023
    any
30 permit                                tcp
    2001:2011::1/64
    any
40 deny                                    any
    any
    any
Hit-counts: enabled
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------

```
---
IPv6      MY_IPV6_ACL
10 permit                                udp
    any
    2001::1/64
20 permit                                tcp
```

```

        2001:2001::2:1                > 1023
        any
30 permit                               tcp
        2001:2011::1/64
        any
40 deny                                any
        any
        any
Hit-counts: enabled

```

Removing an IPv6 ACL:

```

switch(config)# no access-list ipv6 MY_IPV6_ACL
switch(config)# do show access-list
Type      Name
Sequence Comment
Action
Source IP Address      L3 Protocol
Destination IP Address Source L4 Port(s)
Additional Parameters   Destination L4 Port(s)
-----
---
IPv6      MY_IPV6_ACL2
1 permit                               udp
    any
    2001::1/64
2 Permit all TCP ephemeral ports
  permit                               tcp
    2001:2001::2:1                > 1023
    any
3 permit                               tcp
    2001:2011::1/64
    any
4 deny                                any
    any
    any
Hit-counts: enabled

```

Command History

Release	Modification
10.17.1000	Added support for EIGRP, L2TP and VRRP protocols.
10.12	Allow ACLs applied to the Control Plane to be logged.

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<pre>config</pre> The access - list ipv6 <ACL - NAME> command takes you into the named ACL context where you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list log-timer

Syntax

```
access-list log-timer {default|<INTERVAL>}
```

Description

Sets the log timer interval for all ACEs that have the **log** parameter configured.

Parameter	Description
default	Resets the log timer to its default 300 seconds.
<INTERVAL>	Specifies the log timer interval in seconds. Range: 5 to 300.

Usage

- ACL logging keeps a log of the number of packets matching this ACE. Works with both `permit` and `deny` actions. Works with ACLs applied on ingress, or Control Plane, but not with ACLs applied on egress.

- The first packet that matches an ACE with the **log** parameter within an ACL log timer window (configured with the **access-list log-timer** command) has its header contents extracted and sent to the configured logging destination, such as the console and syslog server. Each time the ACL log timer expires, a summary of all ACEs with **log** configured are sent to the logging destination. This capability allows throttling of logging ACL hits.
- If no further log messages are generated in the wait-period, the switch suspends the timer and resets itself to log as soon as a new match occurs.
- When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with the `log` option is logged. Any packets, matching other ACL types, do not create a log until the log-timer wait-period is over. At the end of the wait-period, a summary log is made of all the ACLs that were matched, regardless of type.
- You may see a minor discrepancy between the ACL logging statistics and the hit counts statistics due to the time required to record the log message.

Examples



NOTE

Although these examples use debug logging, you can alternatively use event logging.

Enabling debug logging for the ACL logging module:

```
switch# debug acl log severity info
switch# show debug
-----
module sub_module severity vlan port ip mac instance vrf
-----
acl acl_log info -----
```

Setting the debug destination to console with the minimum security level of info:

```
switch# debug destination console severity info
switch# show debug destination
-----
show debug destination
-----
CONSOLE:info
```

Setting the access list log-timer to 30 seconds:

```
switch(config)# access-list log-timer 30
switch(config)# do show access-list log-timer
ACL log timer length (frequency): 30 seconds
```

Creating an IPv4 ACL with one entry with the log parameter:

```

switch(config)# access-list ip MY_IP_ACL
switch(config-acl-ip)# deny icmp 1.1.1.1 1.1.1.2 log
switch(config-acl-ip)# do show access-list

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv4	MY_IP_ACL	10	deny		icmp	1.1.1.1		1.1.1.2		Logging: enabled Hit-counts: enabled

Enabling interface 1/1/1 and applying the ACL:

```

switch(config)# interface 1/1/1
switch(config-if)# no shutdown
switch(config-if)# no routing
switch(config-if)# apply access-list ip MY_IP_ACL in
switch(config-if)# do show running-config interface 1/1/1
interface 1/1/1
  no shutdown
  apply access-list ip MY_IP_ACL in
  no routing
  vlan access 1
  exit
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ip MY_IP_ACL in
switch(config-if)# do show running-config interface 1/1/1
interface 1/1/1
  no shutdown
  apply access-list ip MY_IP_ACL in
  no routing
  vlan access 1
  exit

```

Sending packets that will match the ACE and observe the ACL logging message on the console:

```

2017-10-10T20:13:36.044+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|
ACL_LOG|
List MY_IP_ACL, seq# 10 denied icmp 1.1.1.1 -> 1.1.1.2 type 8 code 0,
on vlan 1, port 1/1/1, direction in

```

When the access list log-timer expires, the summary message is printed on the console. The number 30 is the number of packets received during the last access list log-timer window.

```
2017-10-10T20:14:06.051+00:00 ops-switchd[875]: debug|LOG_INFO|AMM|1/5|ACL|
ACL_LOG|
MY_IP_ACL on 1/1/1 (in): 30 10 deny icmp 1.1.1.1 1.1.1.2 log count
```

Resetting the ACL log timer to the default value:

```
switch(config)# access-list log-timer default
```

Command History

Release	Modification
10.12	Allow ACLs applied to the Control Plane to be logged.
10.09	<INTERVAL> parameter range changed to 5 to 300 . Was 30 to 300.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

access-list mac

Syntax

```
access-list mac <ACL-NAME>
no access-list mac <ACL-NAME>
[<SEQUENCE-NUMBER>]
{permit|deny}
{any|<SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|<DST-MAC-ADDRESS>[/<ETHERNET-MASK>]}
{any|arp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|
    ipx-arpa|ipx-non-arpa|is-is|lldp|mpls-multicast|mpls-unicast|q-in-q|
    rbridge|trill|wake-on-lan|<NUMERIC-ETHERTYPE>}
```

```
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count] [log]
no <SEQUENCE-NUMBER>
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment
```

Description

Creates a MAC Access Control List (ACL). The ACL is made of one or more Access Control Entries (ACEs) ordered and prioritized by sequence numbers. The lowest sequence number is the highest prioritized ACE.

The **no** form of this command deletes the entire ACL, or deletes an ACE identified by sequence number, or deletes only the comment from the ACE identified by sequence number.

Parameter	Description
<ACL-NAME>	Specifies the name of this ACL.
<SEQUENCE-NUMBER>	Specifies a sequence number for the ACE. Range: 1 to 4294967295.
{permit deny}	Specifies whether to permit or deny traffic matching this ACE.
comment	Specifies storing the remaining entered text as an ACE comment.
{any <SRC-MAC-ADDRESS>[/<ETHERNET-MASK>]] }	Specifies the source host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword <code>any</code> . You can optionally include the following: <ETHERNET-MASK> - The address bits to mask (xxxx.xxx.xxxx).
{any <DST-MAC-ADDRESS>[/<ETHERNET-MASK>]] }	Specifies the destination host MAC address (xxxx.xxxx.xxxx), OUI, or the keyword <code>any</code> . You can optionally include the following: <ETHERNET-MASK> - The address bits to mask (xxxx.xxx.xxxx).
{any arp appletalk ... wake-on-lan <NUMERIC-ETHERTYPE>}	Specifies the protocol encapsulated in the Ethernet frame. The encapsulated protocol is identified by the EtherType Ethernet field. The EtherType is specified in one of the following three ways: <code>any</code> - any EtherType. <NUMERIC-ETHERTYPE> - the numerical EtherType protocol number. Range: 0x600 to 0xffff. - One of these EtherType protocol name keywords: <code>arp</code>

Parameter	Description
	◦ <code>appletalk</code> ◦ <code>arp</code> ◦ <code>fcoe</code> ◦ <code>fcoe-init</code> ◦ <code>ip</code> ◦ <code>ipv6</code> ◦ <code>ipx-arpa</code> ◦ <code>ipx-non-arpa</code> ◦ <code>is-is</code> ◦ <code>lldp</code> ◦ <code>mpls-multicast</code> ◦ <code>mpls-unicast</code> ◦ <code>q-in-q</code> ◦ <code>rbridge</code> ◦ <code>trill</code> ◦ <code>wake-on-lan</code>
<code>pcp <PCP-VALUE></code>	Specifies 802.1Q QoS Priority Code Point value. Range: 0 to 7.
<code>vlan <VID></code>	Specifies a VLAN ID. The VLAN ID must exist.
 NOTE This parameter cannot be used in any ACL that will be applied to a VLAN.	
<code>count</code>	Keeps the hit counts of the number of packets matching this ACE.
<code>log</code>	

Usage

When using multiple ACL types (IPv4, IPv6, or MAC) with logging on the same interface, the first packet that matches an ACE with `log` option is logged. Until the log-timer wait-period is over, any packets matching other ACL types do not create a log. At the end of the wait-period, the switch creates a summary log all the ACLs that were matched, regardless of type.

Examples

Creating a MAC ACL with four entries:

```
switch(config) # access-list mac MY_MAC_ACL
switch(config-acl-ip) # 10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6
```

```

switch(config-acl-ip)# 20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4
switch(config-acl-ip)# 30 permit any any appletalk vlan 40
switch(config-acl-ip)# 40 deny any any any count
switch(config-acl-ip)# exit
switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
---
MAC           MY_MAC_ACL
              10 permit                      ipv6
              1122.3344.5566/ffff.ffff.0000
              any
              20 permit                      any
              aaaa.bbbb.cccc
              1111.2222.3333
              QoS Priority Code Point: 4
              30 permit                      appletalk
              any
              any
              VLAN: 40
              40 deny                      any
              any
              any
              Hit-counts: enabled

```

Adding a comment to an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 30 comment Permit all vlan-40 tagged Appletalk
traffic
switch(config-acl-ip)# exit
switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
---

```

```

MAC          MY_MAC_ACL
10 permit                    ipv6
    1122.3344.5566/ffff.ffff.0000
    any
20 permit                    any
    aaaa.bbbb.cccc
    1111.2222.3333
    QoS Priority Code Point: 4
30 Permit all vlan-40 tagged Appletalk traffic
    permit                    appletalk
    any
    any
    VLAN: 40
40 deny                    any
    any
    any
    Hit-counts: enabled

```

Removing a comment from an existing MAC ACE:

```

switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-mac)# no 30 comment
switch(config-acl-mac)# exit
switch(config)# do show access-list
Type          Name
Sequence Comment
Action
EtherType
Source MAC Address
Destination MAC Address
Additional Parameters
-----
---
MAC          MY_MAC_ACL
10 permit                    ipv6
    1122.3344.5566/ffff.ffff.0000
    any
20 permit                    any
    aaaa.bbbb.cccc
    1111.2222.3333
    QoS Priority Code Point: 4
30 permit                    appletalk
    any
    any
    VLAN: 1
40 deny                    any

```

```
any
any
Hit-counts: enabled
```

Adding an ACE to an existing MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xffee
switch(config-acl-ip)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	EtherType	Source MAC Address	Destination MAC Address	Additional Parameters
------	------	----------	---------	--------	-----------	--------------------	-------------------------	-----------------------

MAC	MY_MAC_ACL							
		10	permit		ipv6	1122.3344.5566/ffff.ffff.0000		
				any				
		20	permit		any	aaaa.bbbb.cccc		
						1111.2222.3333		
				QoS Priority Code Point: 4				
		30	permit		appletalk			
				any				
				any				
				VLAN: 1				
		35	permit		0xffee			
				any				
						aabb.cc11.1234		
		40	deny		any			
				any				
				any				
				Hit-counts: enabled				

Replacing an ACE in an existing MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL
switch(config-acl-ip)# 35 permit any aabb.cc11.1234 0xeeee
switch(config-acl-ip)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	EtherType
------	------	----------	---------	--------	-----------

Source MAC Address
Destination MAC Address
Additional Parameters

```
-----  
---  
MAC      MY_MAC_ACL  
10 permit                               ipv6  
    1122.3344.5566/ffff.ffff.0000  
    any  
20 permit                               any  
    aaaa.bbbb.cccc  
    1111.2222.3333  
    QoS Priority Code Point: 4  
30 permit                               appletalk  
    any  
    any  
    VLAN: 1  
35 permit                               0xeeee  
    any  
    aabb.cc11.1234  
40 deny                                  any  
    any  
    any  
Hit-counts: enabled
```

Removing an ACE from an MAC ACL:

```
switch(config)# access-list mac MY_MAC_ACL  
switch(config-acl-ip)# no 35  
switch(config-acl-ip)# exit  
switch(config)# do show access-list  
Type      Name  
Sequence Comment  
Action                                          EtherType  
Source MAC Address  
Destination MAC Address  
Additional Parameters  
-----  
---  
MAC      MY_MAC_ACL  
10 permit                               ipv6  
    1122.3344.5566/ffff.ffff.0000  
    any  
20 permit                               any  
    aaaa.bbbb.cccc
```

```

1111.2222.3333
QoS Priority Code Point: 4
30 permit                                appletalk
    any
    any
    VLAN: 1
40 deny                                any
    any
    any
Hit-counts: enabled

```

Removing a MAC ACL:

```

switch(config)# no access-list mac MY_MAC_ACL
switch(config)# do show access-list
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
---
MAC           MY_MAC_ACL2
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-40 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> The <code>access-list m</code> <code>ac <ACL-NAME> com</code> mand takes you into the named ACL context whe re you enter the ACEs.	Administrators or local user group members with execution rights for this command.

access-list resequence

Syntax

```
access-list {ip|ipv6|mac} <ACL-NAME> resequence <STARTING-SEQUENCE-NUMBER> <INCREMENT>
```

Description

Resequences the ACE sequence numbers in an ACL.

Parameter	Description
<code>{ip ipv6 mac}</code>	Specifies the ACL type.
<code><ACL-NAME></code>	Specifies the ACL name.
	Specifies the starting sequence number.

Parameter	Description
<STARTING-SEQUENCE-NUMBER>	
<INCREMENT>	Specifies the sequence number increment.

Examples

Resequencing an IPv4 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ip MY_IP_ACL resequence 1 1
switch(config-acl-ip)# exit
switch(config)# do show access-list
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv4	MY_IP_ACL	1	permit		udp	any		172.16.1.0/255.255.255.0		
		2	permit		tcp	172.16.2.0/255.255.0.0	> 1023	any		
		3	permit		tcp	172.26.1.0/255.255.255.0		any		
				dscp: AF11						
				ack						
				syn						
		4	deny		any	any		any		
										Hit-counts: enabled

Resequencing an IPv6 ACL to start at 1 with an increment of 1:

```
switch(config)# access-list ipv6 MY_IPV6_ACL resequence 1 1
switch(config-acl-ip)# exit
```

```
switch(config)# do show access-list
```

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv6	MY_IPV6_ACL		
1	permit		udp
	any		
	2001::1/64		
2	Permit all TCP ephemeral ports		
	permit		tcp
	2001:2001::2:1		> 1023
	any		
3	permit		tcp
	2001:2011::1/64		
	any		
4	deny		any
	any		
	any		
	Hit-counts: enabled		

Type	Name		
Sequence	Comment	Action	L3 Protocol
		Source IP Address	Source L4 Port(s)
		Destination IP Address	Destination L4 Port(s)
		Additional Parameters	

IPv6	MY_IPV6_ACL		
1	permit		udp
	any		
	2001::1/64		
2	Permit all TCP ephemeral ports		
	permit		tcp
	2001:2001::2:1		> 1023
	any		
3	permit		tcp
	2001:2011::1/64		
	any		

```

4 deny                                any
  any
  any
Hit-counts: enabled

```

Resequencing a MAC ACL to start at 1 with an increment of 1:

```

switch(config)# access-list mac MY_MAC_ACL resequence 1 1
switch(config-acl-mac)# exit
switch(config)# do show access-list

```

```

Type      Name
Sequence  Comment
          Action                      EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
---
MAC       MY_MAC_ACL
1 permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
2 permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
  QoS Priority Code Point: 4
3 Permit all vlan-40 tagged Appletalk traffic
  permit                                appletalk
  any
  any
  VLAN: 1
4 deny                                any
  any
  any
Hit-counts: enabled

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

access-list reset

Syntax

```
access-list {all|ip <ACL-NAME>|ipv6 <ACL-NAME>|mac <ACL-NAME>} reset
```

Description

Changes the user-specified ACL configuration to match the active ACL configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Parameter	Description
<code>all ip <ACL-NAME> ipv6 <ACL-NAME> mac <ACL-NAME></code>	Specifies one of the following: • a reset of <code>all</code> ACLs. • a reset of a named IPv4 ACL. • a reset of a named IPv6 ACL. • a reset of a named MAC ACL.

Usage

The output of the **show access-list** command displays the active configuration of the product. The active configuration is the ACLs that have been configured and accepted by the system. The output of the **show access-list** command with the **configuration** parameter, displays the ACLs that have been configured. The output of this command may not be the same as what was programmed in hardware or what is active on the product.

If the active ACLs and user-configured ACLs are not the same, a warning message is displayed in the output of the show command. Modify the user-configured ACL until the warning message is no longer displayed or run the **access-list reset** command to change the user-specified configuration to match the active configuration.

Examples

Apply an ACL with TCP acknowledgments (ACKs) on ingress, which is unsupported by hardware:

```
switch(config-acl)# 10 permit tcp 172.16.2.0/16 any ack
```

Displaying the user-specified configuration:

```
switch(config)# do show access-list commands
! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in
switch(config)# do show access-list commands configuration
! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
access-list ip TEST_ACL
    10 permit tcp 172.16.2.0/255.255.0.0 any ack
! access-list ip TEST_ACL user configuration does not match active
configuration.
! run 'access-list TYPE NAME reset' to reset access-list to match active
configuration.
interface 1/1/1
    apply access-list ip TEST_ACL in
switch(config)# do show access-list
Type          Name
Sequence Comment
              Action          L3 Protocol
              Source IP Address Source L4 Port(s)
              Destination IP Address Destination L4 Port(s)
              Additional Parameters
-----
---
% Warning: TEST_ACL user configuration does not match active configuration.
%          run 'access-list TYPE NAME reset' to reset access-list to match
active configuration.
IPv4          TEST_ACL
switch(config)# do show access-list configuration
Type          Name
```

Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

% Warning: TEST_ACL user configuration does not match active configuration.								
% run 'access-list TYPE NAME reset' to reset access-list to match active configuration.								
IPv4	TEST_ACL							
	10	permit	tcp	172.16.2.0/255.255.0.0				
		any						
		ack						
! access-list ip TEST_ACL user configuration does not match active configuration.								
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.								
access-list ip TEST_ACL								
! access-list ip TEST_ACL user configuration does not match active configuration.								
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.								
interface 1/1/1								
apply access-list ip TEST_ACL in								
switch(config)# do show access-list commands configuration								
! access-list ip TEST_ACL user configuration does not match active configuration.								
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.								
access-list ip TEST_ACL								
	10	permit	tcp	172.16.2.0/255.255.0.0				any ack
! access-list ip TEST_ACL user configuration does not match active configuration.								
! run 'access-list TYPE NAME reset' to reset access-list to match active configuration.								
interface 1/1/1								
apply access-list ip TEST_ACL in								
switch(config)# do show access-list								
Type	Name							
	Sequence	Comment						

```

Action                                     L3 Protocol
Source IP Address                         Source L4 Port(s)
Destination IP Address                     Destination L4 Port(s)
Additional Parameters

-----
---
% Warning: TEST_ACL user configuration does not match active configuration.
%       run 'access-list TYPE NAME reset' to reset access-list to match
active configuration.
IPv4      TEST_ACL
switch(config)# do show access-list configuration
Type      Name
Sequence Comment
Action                                     L3 Protocol
Source IP Address                         Source L4 Port(s)
Destination IP Address                     Destination L4 Port(s)
Additional Parameters

-----
---
% Warning: TEST_ACL user configuration does not match active configuration.
%       run 'access-list TYPE NAME reset' to reset access-list to match
active configuration.
IPv4      TEST_ACL
          10
          permit                               tcp
          172.16.2.0/255.255.0.0
          any
          ack

```

Resetting the user-specified configuration to match the active configuration.

```
switch(config)# access-list ip TEST_ACL reset
```

Displaying the updated user-specified configuration.

```

switch(config)# do show access-list commands
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL in
switch(config)# do show access-list commands configuration
access-list ip TEST_ACL
interface 1/1/1
    apply access-list ip TEST_ACL in
switch(config)# do show access-list
Type      Name
Sequence Comment

```

	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	TEST_ACL	
switch(config)# do show access-list configuration		
Type	Name	
Sequence	Comment	
	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	TEST_ACL	

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

access-list secure-update

Syntax

```
access-list secure-update
no access list secure-update
```

Description

This command determines if access lists are updated using the secure-update feature. Secure-update is enabled by default.

When secure update is enabled and an ACL is updated or replaced, one or more override entries are installed in the TCAM table(s) containing the ACL that is being modified. As a result, all traffic of the same type as the currently configured ACL will be denied on the interfaces to which the ACL is applied. This ensures that traffic is not temporarily allowed while modifying an ACL. Upon completion of the update, the TCAM override entries are uninstalled and traffic resumes ACL matching.

The **no** version of this command disables this feature. If secure-update is disabled, there will be no override entry installed. This results in the faster modification of an ACL and ensures that there is no interruption to previously permitted traffic, but may temporarily allow previously denied traffic to pass through the switch. Once the ACL has been modified, traffic will be processed by the updated ACL.

Examples

Disabling secure update:

```
switch(config)# no access-list secure-update
```

Reenabling secure update:

```
switch(config)# access-list secure-update
```

Related Commands

Command	Description
<code>vsx-sync acl-secure-update</code>	If this setting is enabled and the primary VSX node has configurations with the access list secure - update feature enabled, this configuration can synchronize to the secondary peer. This setting is disabled by default. Refer to the VSX Guide for details.

Command History

Release	Modification
10.13	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

ACL application

ACLs can be applied as follows:

ACL type Direction	IPv4+6 In	MAC In
L2 interface (port)	Yes	Yes
L2 LAG	Yes	Yes
VLAN	Yes	Yes
Control Plane (default VRF)	Yes	



NOTE

The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

TTL on IP ACLs

apply access-list control-plane

Syntax

```
apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>  
no apply access-list {ip|ipv6} <ACL-NAME> control-plane vrf <VRF-NAME>
```

Description

Applies an ACL to the specified VRF.

The **no** form of this command removes application of the ACL from the specified VRF.

Parameter	Description
ip ipv6	Specifies the ACL type: <code>ip</code> for IPv4, or <code>ipv6</code> for IPv6.

Parameter	Description
<code><ACL-NAME></code>	Specifies the ACL name.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.

Usage

Only one ACL per type (**ip**, or **ipv6**) may be applied to a Control Plane VRF at a time. Therefore, using the **apply access-list control-plane** command on a VRF with an already-applied ACL of the same type, will replace the applied ACL.

Examples

Applying **My_ip_ACL** to Control Plane traffic on the default VRF:

```
switch(config) # apply access-list ip My_ip_ACL control-plane vrf default
```

Replacing **My_ip_ACL** with **My_Replacement_ACL** on the default VRF:

```
switch(config) # apply access-list ip My_Replacement_ACL control-plane vrf default
```

Remove (unapply) the **My_Replacement_ACL** from the default VRF. Any other interfaces or VLANs with **My_Replacement_ACL** applied are unaffected.

```
switch(config) # no apply access-list ip My_Replacement_ACL control-plane vrf default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

apply access-list (to interface or LAG)

Syntax

```
apply access-list {ip | ipv6 | mac} <ACL-NAME> {in|out}
no apply access-list {ip | ipv6 | mac} <ACL-NAME> {in | out}
```

Description

Applies an ACL to the interface (Individual front plane port) or Link Aggregation Group (LAG) identified by the current interface or LAG context.

The **no** form of this command removes application of the ACL from the current interface or LAG identified by the current interface or LAG context.

Parameter	Description
ip ipv6 mac	Specifies the ACL type: <code>ip</code> for IPv4, <code>ipv6</code> for IPv6, or <code>mac</code> for MAC ACL.
<ACL-NAME>	Specifies the ACL name.
in	Selects the inbound (ingress) traffic direction.
out	Selects the outbound (egress) traffic direction.



NOTE

For HPE Aruba Networking 6000 and 6100 Switch series, the outbound (egress) traffic direction is supported only for MAC ACLs.

Usage

- Each ACL of a given type can be applied to the same interface or LAG once in each direction. Therefore, using the **apply access-list** command on an interface or LAG with an already-applied ACL of the same type and direction will replace the currently applied ACL.
- An ACL can be applied to an individual front plane port or to a Link Aggregation Group (LAG).
- A port that is a member of a LAG with an applied ACL cannot have a different ACL applied to that member port.

- When the port membership of a LAG with an applied ACL is changed, the LAG ACL is automatically applied or removed from that port depending on the modification type.
- Logging on outbound (egress) traffic is also not supported. An error message will be displayed if an ACL with any logging ACEs is attempted to be applied on egress.

Examples

Applying **My_IP_ACL** to ingress traffic on interface range 1/1/10 to 1/1/12:

```
switch(config)# int 1/1/10-1/1/12
switch(config-if-<1/1/10-1/1/12>)# apply access-list ip My_IP_ACL in
switch(config-if-<1/1/10-1/1/12>)# exit
```

```
switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ip MY_IP_ACL in
switch(config-lag-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_IP_ACL out
switch(config-if)# exit
switch(config)#
```

Applying **MY_IPV6_ACL** to ingress traffic on interface 1/1/1 and to ingress traffic on LAG 100:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-if)# exit
switch(config)# interface lag 100
switch(config-lag-if)# apply access-list ipv6 MY_IPV6_ACL in
switch(config-lag-if)# exit
switch(config)#
```

Applying **MY_MAC_ACL** to ingress traffic on interface 1/1/1 and ingress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL in
switch(config-if)# exit
switch(config)#
```

Applying **MY_MAC_ACL** to egress traffic on interface 1/1/1 and egress traffic on interface 1/1/2:

```
switch(config)# interface 1/1/1
switch(config-if)# apply access-list mac MY_MAC_ACL out
switch(config-if)# exit
switch(config)# interface 1/1/2
switch(config-if)# apply access-list mac MY_MAC_ACL out
```

```

switch(config-if)# exit
switch(config)#
switch(config)# interface 1/1/2
switch(config-if)# apply access-list ip MY_REPLACEMENT_ACL out
switch(config-if)# exit
switch(config)#
switch(config)# interface 1/1/2
switch(config-if)# no apply access-list ip MY_REPLACEMENT_ACL out
switch(config-if)# exit
switch(config)#

```

Command History

Release	Modification
10.12.1000	Egress support extended to 6000 and 6100 Switch series for MAC ACLs.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if config-lag-if	Administrators or local user group members with execution rights for this command.

apply access-list (to VLAN)

Syntax

```

apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}
no apply access-list {ip|ipv6|mac} <ACL-NAME> {in|out}
apply access-list {ip|ipv6|mac} <ACL-NAME> in
no apply access-list {ip|ipv6|mac} <ACL-NAME> in

```

Description

Applies an ACL to the VLAN identified by the current VLAN context.

The **no** form of this command removes application of the ACL from the VLAN identified by the current VLAN context.

Parameter	Description
ip ipv6 mac	Specifies the ACL type: ip for IPv4, ipv6 for IPv6, or mac for MAC ACL.
<ACL-NAME>	Specifies the ACL name.
in	Selects the inbound (ingress) traffic direction.

Selects the outbound (egress) traffic direction.



NOTE

For HPE Aruba Networking 6000 and 6100 Switch series, the outbound (egress) traffic direction is supported only for MAC ACLs.

Usage

- Each ACL of a given type can be applied to the same VLAN once. Therefore, using the **apply access-list** command on a VLAN with an already-applied ACL of the same type, will replace the applied ACL.
- When an ACL is applied to a VLAN, it will create hardware entries on all stack members regardless of whether a VLAN member exists on any specific stack member.

Examples

Applying My_ip_ACL to ingress traffic on VLAN range 20 to 25:

```
switch(config)# vlan 20-25
switch(config-vlan-<20-25>)# apply access-list ip My_ip_ACL in
```

Applying My_ip_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ip My_ip_ACL in
```

Applying My_ipv6_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_ipv6_ACL in
```

Applying My_mac_ACL to ingress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list mac My_mac_ACL in
```

Applying My_mac_ACL to egress traffic on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list mac My_mac_ACL out
```

Replacing My_ipv6_ACL with My_Replacement_ACL on VLAN 10 (following the preceding examples):

```
switch(config)# vlan 10
switch(config-vlan-10)# apply access-list ipv6 My_Replacement_ACL in
```

Removing (unapplying, Specifies the ACL type: **ip** for IPv4, **ipv6** for IPv6, or **mac** for MAC ACL.) several ACLs on VLAN 10:

```
switch(config)# vlan 10
switch(config-vlan-10)# no apply access-list ipv6 My_Replacement_ACL in
switch(config-vlan-10)# no apply access-list mac My_mac_ACL in
```

Command History

Release	Modification
10.12.1000	Egress support extended to 6000 and 6100 Switch series for MAC ACLs.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-vlan	Administrators or local user group members with execution rights for this command.

clear access-list hitcounts

Syntax

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>]
                             [interface <IF-NAME>| vlan <VLAN-ID>] [in] }
```

```
clear access-list hitcounts { all | [{ip|ipv6|mac} <ACL-NAME>]  
                             [interface <IF-NAME>| vlan <VLAN-ID>] [in|out] }
```

Description

Clears the hit counts for ACLs with ACEs that include the `count` keyword.

Parameter	Description
<code>all</code>	Selects all ACLs.
<code>ip ipv6 mac</code>	Specifies the ACL type: <code>ip</code> for IPv4, <code>ipv6</code> for IPv6, or <code>mac</code> for MAC.
<code><ACL-NAME></code>	Specifies the ACL name.
<code>interface <IF-NAME></code>	Specifies the interface name (port or LAG).
<code>vlan <VLAN-ID></code>	Specifies the VLAN.
<code>in</code>	Selects the inbound (ingress) traffic direction.
<code>out</code>	Selects the outbound (egress) traffic direction.



NOTE

For HPE Aruba Networking 6000 and 6100 Switch series, the outbound (egress) traffic direction is supported only for MAC ACLs.

Examples

```
switch# clear access-list hitcounts ip My_ip_ACL interface 1/1/2 out
```

Clearing the hit counts for My_ip_ACL applied to VLAN 10 (ingress):

```
switch# clear access-list hitcounts ip My_ip_ACL vlan 10 in
```

Clearing the hit counts for My_mac_ACL applied to VLAN 10 (egress):

```
switch# clear access-list hitcounts mac My_mac_ACL vlan 10 out
```

Clearing the hit counts for all ACLs:

```
switch# clear access-list hitcounts all
```

Command History

Release	Modification
10.12.1000	Egress support extended to 6000 and 6100 Switch series for MAC ACLs.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

clear access-list hitcounts control-plane

Syntax

```
clear access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

Description

Clears the hit counts for ACLs applied to the Control Plane VRF.

Parameter	Description
ip ipv6	Specifies the ACL type: ip for IPv4, or ipv6 for IPv6.
<ACL-NAME>	Specifies the ACL name.

Parameter	Description
vrf <VRF-NAME>	Specifies the VRF name.

Examples

Clearing the hit counts for an IPv4 ACL applied to the Control Plane `default` VRF:

```
switch# clear access-list hitcounts ip My_ipv4_ACL control-plane vrf default
```

Clearing the hit counts for an IPv6 ACL applied to the Control Plane `default` VRF:

```
switch# clear access-list hitcounts ipv6 My_ipv6_ACL control-plane vrf
default
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (<code>></code>) or Manager (<code>#</code>)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

show access-list

Syntax

10.05 reviewer comment: Added routed-in and routed-out for FL, GL, ML, TL and routed-in only for XL. Removed exclusion of IPv4/IPv6 egress VLANs for GL and TL. 10.05 May 28 2020: SME Kyle Klockenbrink (Julie). Added routed-in for FL GL TL XL and routed-out for FL GL TL (NOT XL). ML gains NEITHER routed-in or routed-out.

```
show access-list [interface <IF-NAME> | vlan <VLAN-ID> | interface
vlan <VLAN-ID> | interface lag <LAG-ID>] [ip|ipv6|mac] [in] [commands]
[configuration]
show access-list [interface <IF-NAME> | vlan <VLAN-ID> | interface vlan
<VLAN-ID> | interface lag <LAG-ID>] [ip|ipv6|mac] [in|out] [commands]
[configuration]
show access-list [ip|ipv6|mac] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]
[commands] [configuration]
```

Description

Shows information about your defined ACLs and where they have been applied. When **show access-list** is entered without parameters, information for all ACLs is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL.
- All ACLs of a specific type.
- The ACL applied to a specific interface (port or split port or LAG).
- The ACL applied to a specific VLAN.
- All IPv4 or IPv6 ACLs applied to interface VLANs.

Parameter	Description
interface <IF-NAME>	Specifies the interface name (port or LAG).
vlan <VLAN-ID>	Specifies the VLAN.
in	Selects the inbound (ingress) traffic direction.
out	Selects the outbound (egress) traffic direction.
 NOTE For HPE Aruba Networking 6000 and 6100 Switch series, the outbound (egress) traffic direction is supported only for MAC ACLs.	
ip ipv6 mac	Specifies the ACL type: • ip for IPv4, • ipv6 for IPv6, or

Parameter	Description
	mac for MAC.
<ACL-NAME>	Specifies the ACL name.
commands	Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.
configuration	Specifies that the user - configured ACLs be shown as entered, even if the ACLs are not active due to ACE - definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

Examples

Creating an IPv4 ACL, applying it to an interface VLAN (routed in), and then showing ACL information filtered for that interface VAN:

```
switch(config)# access-list ip test
switch(config-acl-ip)# 10 permit any 1.1.1.2 2.2.2.2 count
switch(config-acl-ip)# 20 permit any 1.1.1.2 2.2.2.1 count
switch(config-acl-ip)# 30 permit any 2.2.2.2 1.1.1.2 count
switch(config-acl-ip)# 40 permit any 2.2.2.2 1.1.1.1 count
switch(config-acl-ip)# 50 permit any any any count
switch(config-acl-ip)# exit
switch(config)#
switch(config)# interface vlan100
switch(config-if-vlan)# apply access-list ip test routed-in
switch(config-if-vlan)# exit
switch(config)# show access-list interface vlan100 ip routed-in
Direction
Type      Name
Sequence Comment
          Ac  L3 Protocol
          Source IP Address      Source L4 Port(s)
          Destination IP Address Destination L4 Port(s)
          Additional Parameters
-----
---
Routed Inbound
IPv4      test
          10
```

```

        permit                                any
        1.1.1.2
        2.2.2.2
        Hit-counts: enabled
20
        permit                                any
        1.1.1.2
        2.2.2.1
        Hit-counts: enabled
30
        permit                                any
        2.2.2.2
        1.1.1.2
        Hit-counts: enabled
40
        permit                                any
        2.2.2.2
        1.1.1.1
        Hit-counts: enabled
50
        permit                                any
        any
        any
        Hit-counts: enabled

```

```

-----
---
```

Showing an IPv4 ACL:

```
switch# show access-list ip MY_ACL
```

```

Type      Name
Sequence  Comment
          Action                                L3 Protocol
          Source IP Address                    Source L4 Port(s)
          Destination IP Address                Destination L4 Port(s)
          Additional Parameters
-----
--
IPv4      MY_ACL
10 permit                                udp
   any
   172.16.1.0/255.255.255.0
20 permit                                tcp
   172.16.2.0/255.255.0.0                > 1023
   any

```

```

30 permit                                tcp
   172.26.1.0/255.255.255.0
   any
   syn
   ack
   dscp 10
40 deny                                any
   any
   any
Hit-counts: enabled
-----
--

```

Showing an IPv4 ACL as commands:

```

switch# show access-list ip MY_ACL commands
access-list ip MY_ACL
  10 permit udp any 172.16.1.0/255.255.255.0
  20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any
  30 permit tcp 172.26.1.0/255.255.255.0 any syn ack dscp 10
  40 deny any any any any count

```

```

switch# show access-list interface 1/1/2.1 mac in

```

Direction

Type	Name
Sequence	Comment
Action	EtherType
Source MAC Address	
Destination MAC Address	
Additional Parameters	

```

-----
---
Inbound
MAC      My_mac_ACL
10
  permit                                ipv6
  1122.3344.5566/ffff.ffff.0000
  any
20
  permit                                any
  aaaa.bbbb.cccc
  1111.2222.3333
  QoS Priority Code Point: 4
30
  deny                                any
  any

```

```
any
Hit-counts: enabled
```

Showing IPv4 ACLs applied to VLAN 10, inbound:

```
switch# show access-list vlan 10 ip in
```

Type	Name		
Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		
IPv4	My_ip_ACL		
10	permit		udp
	any		
	172.16.1.0/255.255.255.0		
20	permit		tcp
	172.16.2.0/255.255.0.0		> 1023
	any		
30	permit		tcp
	172.26.1.0//255.255.255.0		
	any		
	syn		
	ack		
	dscp 10		
40	deny		any
	any		
	any		
	Hit-counts: enabled		

Showing IPv6 ACLs applied to LAG 128, inbound:

```
switch# show access-list interface lag128 ipv6 in
```

Type	Name		
Sequence	Comment		
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		

```

IPv6      MY_IPV6_ACL
10 permit                                udp
    any
    2001::1/64
20 permit                                tcp
    2001:2001::2:1/128                > 1023
    any
30 permit                                tcp
    2001:2011::1/64
40 deny                                    any
    any
    any
    Hit-counts: enabled
-----
--

```

Showing an IPv6 ACL as commands:

```

switch# show access-list ipv6 MY_IPV6_ACL commands
access-list ipv6 MY_IPV6_ACL
  10 permit udp any 2001::1/64
  20 permit tcp 2001:2001::2:1/128 gt 1023 any
  40 deny any any any count

```

Showing a MAC ACL:

```

switch# show access-list mac MY_MAC_ACL
Type      Name
Sequence  Comment
          Action                                EtherType
          Source MAC Address
          Destination MAC Address
          Additional Parameters
-----
--
MAC       MY_MAC_ACL
10 permit                                ipv6
    1122.3344.5566/ffff.ffff.0000
    any
20 permit                                any
    aaaa.bbbb.cccc
    1111.2222.3333
    QoS Priority Code Point: 4
30 deny                                    any
    any
    any
    Hit-counts: enabled

```

```
-----  
--
```

Showing a MAC ACL as commands:

```
switch# show access-list mac MY_MAC_ACL commands  
access-list mac MY_MAC_ACL  
  10 permit 1122.3344.5566/ffff.ffff.0000 any ipv6  
  20 permit aaaa.bbbb.cccc 1111.2222.3333 any pcp 4  
  30 deny any any any count
```

Command History

Release	Modification
10.12.1000	Egress support extended to 6000 and 6100 Switch series for MAC ACLs.
10.07 or earlier	Command introduced.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show access-list control-plane

Syntax

```
show access-list [ip|ipv6] [<ACL-NAME>] control-plane [vrf <VRF-NAME>]  
                  [commands] [configuration]
```

Description

Shows information about your defined ACLs that have been applied to the Control Plane. When **show access-list control-plane** is entered without parameters, information for all ACLs applied to the Control Plane is shown. The parameters filter the list of ACLs for which information is shown.

Available filtering includes:

- The content of a specific ACL that has been applied to the Control Plane.

- All ACLs of a specific type that have been applied to the Control Plane.
- All ACLs applied to the Control Plane for a specific VRF.

Parameter	Description
<code>ip ipv6</code>	Specifies the ACL type: <code>ip</code> for IPv4, or <code>ipv6</code> for IPv6.
<code><ACL-NAME></code>	Specifies the ACL name.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.
<code>[commands]</code>	Specifies that the ACL definition is to be shown as the commands and parameters used to create it rather than in tabular form.
<code>[configuration]</code>	Specifies that the user - configured ACLs be shown as entered, even if the ACLs are not active due to ACE - definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previous successfully programmed (active) ACLs configuration.

Examples

Showing an IPv4 ACL applied to the Control Plane `default` VRF:

```
switch# show access-list ip My_ipv4_ACL control-plane vrf default
Type      Name
Sequence  Comment
          Action                    L3 Protocol
          Source IP Address          Source L4 Port(s)
          Destination IP Address      Destination L4 Port(s)
          Additional Parameters
-----
--
IPv4      My_ipv4_ACL
          10 permit                    udp
              any
              172.16.1.0/24
          20 permit                    tcp
              172.16.2.0/16          > 1023
              any
          30 permit                    tcp
```

```

172.26.1.0/24
any
syn
ack
dscp 10
40 deny any
any
any
Hit-counts: enabled
-----
--

```

Showing an IPv6 ACL applied to the Control Plane `default` VRF:

```
switch# show access-list ipv6 My_ipv6_ACL control-plane vrf default
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
IPv6	My_ipv6_ACL	10	permit		udp	any		2001::1/64		
		20	permit		tcp	2001:2001::2:1/128	> 1023	any		
		30	permit		tcp	2001:2011::1/64				
		40	deny		any	any		any		
										Hit-counts: enabled

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show access-list hitcounts

Syntax

```
show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> | vlan <VLAN-ID>] [in]]}
show access-list hitcounts { [{ip|ipv6|mac} <ACL-NAME>] [interface <IF-NAME> | vlan <VLAN-ID>] [in|out]]}
```

Description

Shows the hit count of the number of times an ACL has matched a packet or frame for ACEs with the **count** keyword. For ACEs without the **count** keyword, a dash is shown in place of a hit count.

Parameter	Description
ip ipv6 mac	Specifies the ACL type: ip for IPv4, ipv6 for IPv6, or mac for MAC.
<ACL-NAME>	Specifies the ACL name.
interface <IF-NAME>	Specifies the interface name (port or split port or LAG).
vlan <VLAN-ID>	Specifies the VLAN.
in	Selects the inbound (ingress) traffic direction.
out	Selects the outbound (egress) traffic direction.



NOTE

Parameter

Description



For HPE Aruba Networking 6000 and 6100 Switch series, the outbound (egress) traffic direction is supported only for MAC ACLs.

Usage

- ACL hit counts are aggregated across all:
 - Physical interfaces to which the ACL is applied to on ingress.
 - VLANs to which the ACL is applied to on ingress.
- If an ACL with an ACE with the **count** keyword is applied to multiple physical interfaces or VLANs, the hit counts are aggregated. There is one aggregation for physical interfaces and another for VLANs.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for **My_ip_ACL** applied to port 1/1/2:

```
switch# show access-list hitcounts ip My_ip_ACL interface 1/1/2
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/1-1/1/2,lag1 (out):
    Matched Packets  Configuration
        - 10 permit udp any 172.16.1.0/255.255.255.0
        0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
        - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
        0 implicit deny any any any count
```

Showing the hit counts for **My_ip_ACL** applied to VLAN 10:

```
switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
vlan 10,20-100,300 (in):
    Matched Packets  Configuration
        - 10 permit udp any 172.16.1.0/255.255.255.0
        0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
        - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
        0 implicit deny any any any count

switch# show access-list hitcounts ip My_ip_ACL interface 1/1/4.1
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/4.1,1/1/10.10 (in):
    Matched Packets  Configuration
```

```

- 10 permit udp any 172.16.1.0/255.255.255.0
0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
- 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
0 implicit deny any any any count
switch# show access-list hitcounts ip My_ip_ACL2 interface lag1.3
Statistics for ACL My_ip_ACL2 (ipv4):
interface lag1.3-lag1.4 (in):
    Matched Packets Configuration
        0 10 deny icmp any 192.168.42.1 count
    3884 100 permit any any any count
        0 implicit deny any any any count

```

Showing the hit counts for **My_ip_ACL** applied to interface VLAN 10:

```

switch# show access-list hitcounts ip My_ip_ACL vlan 10
Statistics for ACL My_ip_ACL (ipv4):
interface vlan 10,20,30 (routed-in):
    Matched Packets Configuration
        - 10 permit udp any 172.16.1.0/255.255.255.0
        0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
        - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
0 implicit deny any any any count

```

Showing the hit counts for **My_ip_ACL** applied on any interface and direction:

```

switch# show access-list hitcounts ip My_ip_ACL vlan 10
switch# show access-list hitcounts ip My_ip_ACL
Statistics for ACL My_ip_ACL (ipv4):
interface 1/1/1 (in):
    Matched Packets Configuration
        - 10 permit udp any 172.16.1.0/255.255.255.0
        0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
        - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
0 implicit deny any any any count

interface 1/1/4.1,1/1/10.10 (in):
    Matched Packets Configuration
        - 10 permit udp any 172.16.1.0/255.255.255.0
        0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
        - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
0 implicit deny any any any count

interface vlan 10,20,30 (routed-in):

```

```

    Matched Packets Configuration
      - 10 permit udp any 172.16.1.0/255.255.255.0
    0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
      - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
    0 implicit deny any any any count
interface vlan 80-85 (routed-out):
    Matched Packets Configuration
      - 10 permit udp any 172.16.1.0/255.255.255.0
    0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
      - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
    0 implicit deny any any any count
vlan 10,20-100,300 (in):
    Matched Packets Configuration
      - 10 permit udp any 172.16.1.0/255.255.255.0
    0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
      - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
    0 implicit deny any any any count

vrf blue,default,red (control-plane):
    Matched Packets Configuration
      - 10 permit udp any 172.16.1.0/255.255.255.0
    0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
      - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
    0 implicit deny any any any count

```

Command History

Release	Modification
10.12.1000	Egress support extended to 6000 and 6100 Switch series for MAC ACLs.
10.07 or earlier	Updated command output to use interface and VLAN ranges to reflect aggregation.

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show access-list hitcounts control-plane

Syntax

```
show access-list hitcounts [{ip|ipv6} <ACL-NAME>] control-plane vrf <VRF-NAME>
```

Description

Shows the hit count of the number of times an ACL (applied to the Control Plane) has matched a packet for ACEs with the `count` keyword. For ACEs without the `count` keyword, a dash is shown in place of a hit count.

Parameter	Description
<code>ip ipv6</code>	Specifies the ACL type: ip for IPv4, or ipv6 for IPv6.
<code><ACL-NAME></code>	Specifies the ACL name.
<code>vrf <VRF-NAME></code>	Specifies the VRF name.

Usage

- ACL hit counts are aggregated across all VRFs to which the ACL is applied to on ingress.
- Accumulated hit counts for an applied ACL are cleared upon any modification of the ACL.

Examples

Showing the hit counts for an IPv4 ACL applied to the Control Plane default VRF:

```
switch# show access-list hitcounts ip My_ipv4_ACL control-plane vrf default
Statistics for ACL My_ip_ACL (ipv4):
vrf default (control-plane):
    Matched Packets  Configuration
                    - 10 permit udp any 172.16.1.0/255.255.255.0
                    0 20 permit tcp 172.16.2.0/255.255.0.0 gt 1023 any count
                    - 30 permit tcp 172.26.1.0/255.255.255.0 any dscp AF11
ack syn
                    0 implicit deny any any any count
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show access-list secure-update

Syntax

```
show access-list secure-update
```

Description

Use this command to determine if access lists are updated using the secure-update feature. Secure update is enabled by default.

Examples

Displaying the status of the access list secure-update feature when that feature is enabled:

```
switch(config)# show access-list secure-update
Access-list secure-update is enabled
```

Displaying the status of the access list secure-update feature when that feature is disabled:

```
switch(config)# show access-list secure-update
Access-list secure-update is disabled
```

Related Commands

Command	Description
<code>access-list secure-update</code>	This command determines if access lists are updated using the secure - update feature. Secure - update is enabled by default.

Command History

Release	Modification
10.13	Command introduced

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code>	Administrators or local user group members with execution rights for this command.

show capacities

Syntax

```
show capacities <FEATURE>
```

Description

Shows system capacities and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies a feature. For example, aaa .

Usage

Capacities are expressed in user-understandable terms. Thus they may not map to a specific hardware or software resource or component. They are not intended to define a feature exhaustively.

Examples

Showing classifier-related capacities on the 4100i, 6000, or 6100:

```
switch# show capacities classifier
System Capacities: Filter Classifier
Capacities Name
Value
-----
-----
Maximum number of Access Control Entries configurable in a
system                      4096
Maximum number of Access Control Lists configurable in a
system                      512
Maximum number of class entries configurable in a
system                      4096
Maximum number of classes configurable in a
system                      512
Maximum number of entries in an Access Control
List                        1024
Maximum number of entries in a
class                      1024
Maximum number of entries in a
policy                      64
Maximum number of classifier policies configurable in a
system                      512
Maximum number of policy entries configurable in a
system                      4096
Showing all available capacities on the 4100i, 6000, or 6100:
switch# show capacities
System Capacities:
Capacities
Name                      Value
-----
-----
```

Maximum number of Access Control Entries configurable in a system	4096
Maximum number of Access Control Lists configurable in a system	512
Maximum number of class entries configurable in a system	4096
Maximum number of classes configurable in a system	512
Maximum number of entries in an Access Control List	1024
Maximum number of entries in a class	1024
Maximum number of entries in a policy	64
Maximum number of classifier policies configurable in a system	512
Maximum number of policy entries configurable in a system	4096
Maximum number of clients supported for tracking the IP address in the system	128
Maximum number of dynamic VLANs that can be allowed using MVRP	256
Maximum number of nexthops per IP ECMP group	1
Maximum number of IP neighbors (IPv4+IPv6) supported in the system	1024
Maximum number of IPv4 neighbors(# of ARP entries) supported in the system	1024
Maximum number of IPv6 neighbors(# of ND entries) supported in the system	512
Maximum number of L2 MAC addresses supported in the system	8192
Maximum number of L3 Groups for IP Tunnels and ECMP Groups	1
Maximum number of L3 Destinations for Routes, Nexthops in Tunnels and ECMP groups	1024
Maximum number of configurable LAG ports	8
Maximum number of members supported by a LAG port	8
Maximum number of VLANs across ports allowed in loop-protect	3328
Maximum number of IGMP/MLD groups	

supported	512
Maximum number of IGMP/MLD snooping groups supported	512
Maximum number of Mirror Sessions configurable in a system	4
Maximum number of enabled Mirror Sessions in a system	4
Maximum number of mstp instances configurable in a system	16
Maximum number of Clients that can be authenticated on a port	32
Maximum number of Device Profiles allowed to be created on the system	8
Maximum number of Port Access Roles allowed to be created on the system	32
Maximum number of MAC Address that can be authorized on a port	32
Maximum number of Port Access Role VLAN IDs allowed to be created on the system	50
Maximum number of Port Access Role VLAN names allowed to be created on the system	50
Maximum number of RBAC rules per user group	1024
Maximum number of RPVST VLANs configurable on the system	16
Maximum number of RPVST VPORTs supported in a system	512
Maximum number of SVIs supported in the system	16
Maximum number of routes (IPv4+IPv6) on the system	512
Maximum number of IPv4 routes on the system	512
Maximum number of IPv6 routes on the system	512
Maximum number of VLANs supported in the system	512

Showing all available capacities for mirroring:

```
switch# show capacities mirroring
System Capacities: Filter Mirroring
Capacities Name
Value
-----
```

```

-----
Maximum number of Mirror Sessions configurable in a
system                                4
Maximum number of enabled Mirror Sessions in a
system                                4

```

Showing all available capacities for MSTP:

```

switch# show capacities mstp
System Capacities: Filter MSTP
Capacities Name
Value
-----
-----
Maximum number of mstp instances configurable in a
system                                64

```

Showing all available capacities for VLAN count:

```

switch# show capacities vlan-count
System Capacities: Filter VLAN Count
Capacities Name
Value
-----
-----
Maximum number of VLANs supported in the
system                                4094
/switch# show capacities
vlan-count
System Capacities: Filter VLAN Count
Capacities Name
Value
-----
-----
Maximum number of VLANs supported in the
system                                4094

```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show capacities-status

Syntax

```
show capacities-status <FEATURE>
```

Description

Shows system capacities status and their values for all features or a specific feature.

Parameter	Description
<FEATURE>	Specifies the feature, for example aaa for which to display capacities, values, and status. Required.

Examples

Showing the system capacities status for all features:

```
switch# show capacities-status
System Capacities Status
Capacities Status Name
Value Maximum
-----
Number of Access Control Entries currently configured
0      4096
Number of Access Control Lists currently configured
0      512
Number of class entries currently configured
0      4096
Number of classes currently configured
```

```

0      512
Number of policies currently configured
0      512
Number of policy entries currently configured
0      4096
Number of dynamic VLANs currently learnt using MVRP
0      256
Number of IP neighbor (IPv4+IPv6) entries
1      1024
Number of IPv4 neighbor(ARP) entries
1      1024
Number of IPv6 neighbor(ND) entries
0      512
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured
0      1
Number of L3 Destinations for Routes, Nexthops in ECMP groups and Tunnels
    currently configured
0      1024
Number of Mirror Sessions currently configured
0      4
Number of Mirror Sessions currently enabled
0      4
Number of mstp instances currently configured
0      16
Number of RPVST VLANs currently configured
0      16
Number of routes (IPv4+IPv6) currently configured
1      512
Number of IPv4 routes currently configured
1      512
Number of IPv6 routes currently configured with prefix 0-64
0      512
Number of IPv6 routes currently configured with prefix 65-127
0      512
Number of VLANs currently configured
1      512

```

Showing the system capacities status for BGP:

Showing the system capacities status for L3:

```

switch# show capacities-status l3
System Capacities Status: Filter L3 resources
Capacities Status Name
Value Maximum
-----

```

```

-----
Number of IP neighbor (IPv4+IPv6) entries
4      49152
Number of IP Directed Broadcast neighbor entries
0      4096
Number of IPv6 Long Prefix Routes currently configured
3      5000
Number of IPv6 neighbor(ND) entries
4      49152
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured
1      2047
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
      Tunnels currently configured
4      2045
Number of routes (IPv4+IPv6) currently configured
5      65536
Number of IPv4 routes currently configured
0      65536
Number of IPv6 routes currently configured with prefix 0-64
4      13312
Number of IPv6 routes currently configured with prefix 65-127
2      510

```

switch# **show capacities-status l3**

System Capacities Status: Filter L3 resources

Capacities Status Name

Value Maximum

```

-----
Number of IP neighbor (IPv4+IPv6) entries
4      49152
Number of IP Directed Broadcast neighbor entries
0      4096
Number of IPv6 Long Prefix Routes currently configured
3      5000
Number of IPv6 neighbor(ND) entries
4      49152
Number of L3 Groups for IP Tunnels and ECMP Groups currently configured
1      2047
Number of L3 Destinations for Routes, Nexthops in ECMP groups and
      Tunnels currently configured
4      2045
Number of routes (IPv4+IPv6) currently configured
5      65536

```

```
Number of IPv4 routes currently configured
0      65536
Number of IPv6 routes currently configured with prefix 0-64
4      13312
Number of IPv6 routes currently configured with prefix 65-127
2      510
```

Command History

Release	Modification
10.13	Updated to show newly supported configuration of IPv6 routes on the ASIC.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Classifier policies

Classifier policies let a network administrator define sets of rules based on network traffic addressing or other header content, and use these rules to restrict or alter the passage of traffic through the switch. Choosing the rule criteria is called Classification, and one such rule, or list, is called a policy. Classification is achieved by creating a traffic class. The types of classes (MAC, IPv4, and IPv6) are each focused on relevant frame/packet characteristics. Classes can be configured to match or ignore almost any frame or packet header field. Network traffic passing through a switch can be classified based on many different frame/packet characteristics including, but not limited to:

- Frame ingress VLAN ID
- Source and/or destination Ethernet MAC, IPv4, or IPv6 address
- Layer 2 (EtherType) and Layer 3 (IP) protocol
- Layer 4 application ports

A policy contains one or more policy entries, which are listed according to priority by sequence number. A single policy entry contains a class and corresponding policy action. Policy action is taken on traffic matched

by its corresponding class. A policy can be applied to an individual front plane port, a Link Aggregation Group (LAG) interface, or a VLAN.

**NOTE**

See also [ACL and Policy Hardware Resource Considerations](#).

Subtopics

Traffic policing

Types of policy actions

How policy matching works

Active class configuration versus user-specified configuration

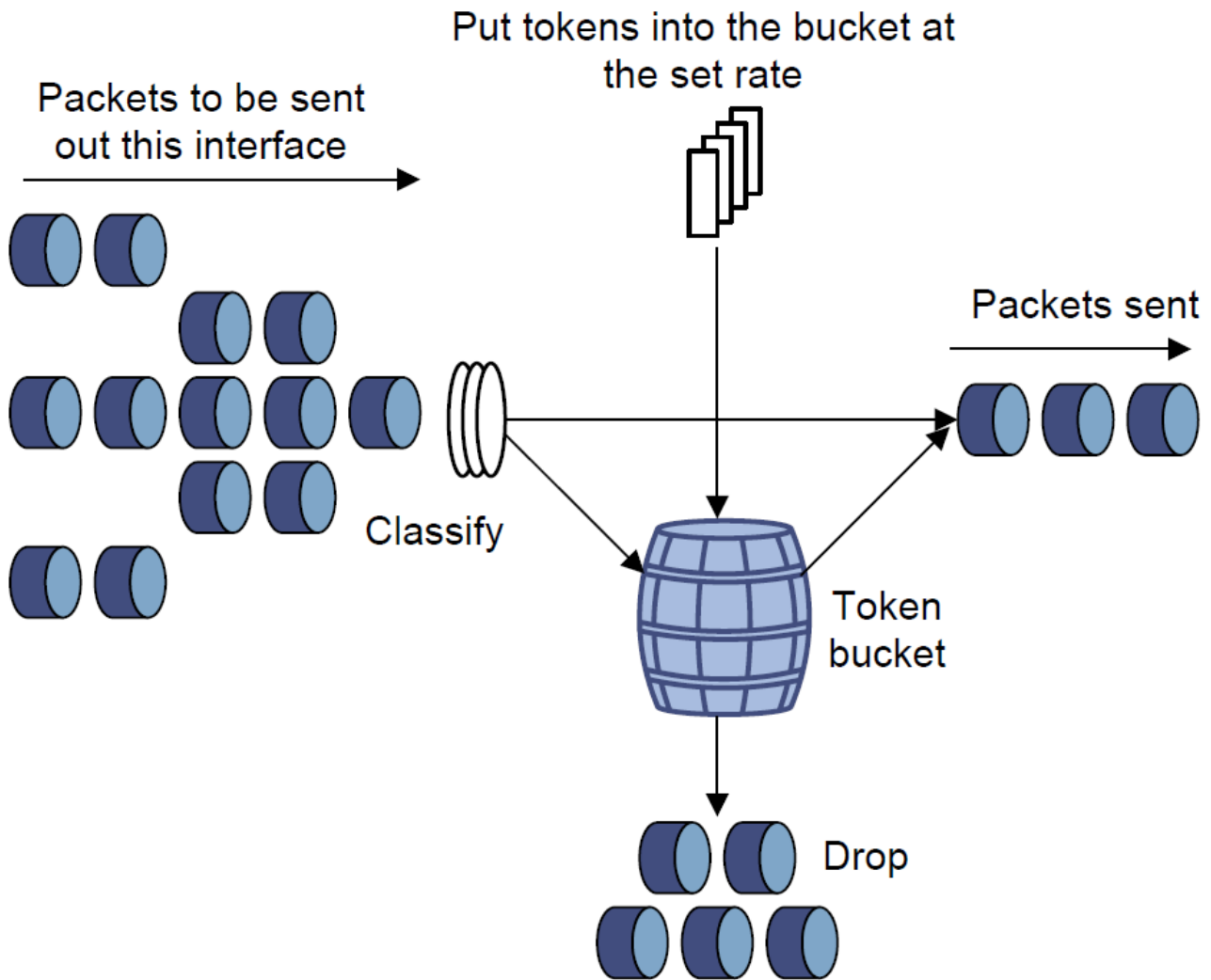
Active policy configuration versus user-specified configuration

Classifier policy application

Classifier policy commands

Traffic policing

Traffic policing supports policing of the inbound traffic. A typical application of traffic policing is to supervise the specification of traffic entering a network and limit it within a reasonable range. Another application is to "discipline" the extra traffic to prevent aggressive use of network resources by an application. For example, you can limit bandwidth for HTTP packets to less than 50% of the total. If the traffic of a session exceeds the limit, traffic policing can drop the packets or reset the IP precedence of the packets. In the following illustrated example, outbound traffic is policed:



Traffic policing is widely used in policing traffic entering the ISP networks. It can classify the policed traffic and take predefined policing actions on each packet depending on the evaluation result:

- Forwarding the packet if the evaluation result is "conforming."
- Dropping the packet if the evaluation result is "excess."
- Forwarding the packet with its precedence remarked when the evaluation result is "conforming."

Types of policy actions

The policy actions are broadly classified in the following categories:

- Remark actions
- Police actions
- Other actions

Each policy entry can have a combination of policy actions from these multiple categories, which are executed in the order of configuration.

Remark actions

This category contains the following actions:

- **IP precedence:** 3-bit field in IP header which denotes the importance or priority of the datagram.
- **IP Differentiated Services Code Point (DSCP):** 6-bit field in IP header for packet classification.
- **Class of Service (CoS):** Queuing can be achieved through the cos action field which will remark the cos value in case of tagged packets and queuing of both tagged and untagged packets.

Police actions

Traffic policing meters inbound traffic on an interface or VLAN based on the following traffic parameters:

- Committed information rate (CIR): Bandwidth limit for guaranteed traffic.

Based on these parameters, packets are dropped when traffic exceeds the bandwidth limit (CIR).

Other actions

Other actions include Drop: Drop the packet, and Mirror: Mirror the packets to a specified mirroring session. For details, see the Monitoring Guide .

How policy matching works

A policy can be applied to an interface or VLAN to affect/control traffic arriving on that interface or VLAN (inbound (ingress)). A single policy entry matches on one or more characteristics of the particular traffic type and has a configured action to continue through the switch. This matching occurs by beginning with the entry with the lowest sequence number. The entry is then compared against the incoming frame to its particular match characteristics. If there is a match, the action is taken.

If there is no match, the match characteristics of the next sequence are compared to the relevant frame/packet details. If there is a match, the specified actions are taken. This process continues until a match is found; otherwise, the packet is permitted to flow through the switch unaltered. The "implicit permit" behavior of policy matching differs from the "implicit deny" behavior of ACL matching.

Active class configuration versus user-specified configuration

Syntax

The output of the `show class` command displays the active class configurations. Active class configurations are the classes that have been configured and accepted by the system.

The output of the `show class` command with the `configuration` parameter, displays the classes that have been configured by the user.

Discrepancies might occur between the active class configurations and the user-specified configurations. In the user-specified class configurations, unsupported command parameters may have been configured, or class can be modified after policy application and may have been unsuccessful due to lack of hardware resources.

To determine if a discrepancy exists between what was configured and what is active, run any variant of the `show class` command. If the active classes and configured classes are not the same, a warning message is displayed.

```
! class MY_CLASS user configuration does not match active configuration.  
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command with parameters, the following in-progress message is displayed:

```
! class ip MY_CLASS user configuration currently being processed  
! run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the configured class is processing and you entered the `show class` command without parameters, the following in-progress message is displayed:

```
% Warning: MY_CLASS user configuration currently being processed  
% run 'class TYPE NAME reset' to reset class to match active configuration.
```

If the warning message or in-progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `class {all|ip <class-name>|ipv6 <class-name> } reset` command to change the user-specified configuration to match the active configuration.



NOTE

The `show running-config` command also shows a warning about classes that are in progress or failed.

Example

Resetting the user-specified class configuration to the active configuration:

```
switch(config) # class all reset
```

Active policy configuration versus user-specified configuration

Syntax

The output of the `show policy` command displays the active policy configurations. Active policy configurations are the policies that have been configured and accepted by the system. With applied policies, the active configuration displays the interfaces on which the policies have successfully been programmed in hardware.

The output of the `show policy` command with the `configuration` parameter, displays the policies that have been configured by the user.

Discrepancies might exist between the active policy configurations and the user-specified configurations. In the user-specified policy configurations, unsupported command parameters might have been configured, or an application of a policy might have been unsuccessful because of a lack of hardware resources.

To determine if a discrepancy exists between the configuration and what is active, run any variant of the `show policy` command. If the active policies and configured policies are not the same, a warning message is displayed in the output of the `show` command.

```
! policy MY_POLICY user configuration does not match active configuration.  
! run 'policy NAME reset' to reset policy to match active configuration.
```

The switch displays an `in progress` message while it processes the configured policy:

```
! policy MY_POLICY user configuration currently being processed  
! run 'policy NAME reset' to reset policy to match active configuration.
```

If the warning message or in progress message is displayed, additional changes may be made until the error message is no longer displayed. Or you can use the `policy <policy-name> reset` command to change the user-specified configuration to match the active configuration.

Example

Resetting `MY_POLICY`:

```
switch(config)# policy MY_POLICY reset
```

Subtopics

Considerations for when a policy is applied per interface

Considerations for when a policy is applied per interface



NOTE

This section is only applicable to policies applied to physical interfaces and LAGs using the `per-interface` parameter.

The `reset` command (mentioned in the previous section) is not useful if one or more unique instances of a policy created using the `per-interface` parameter fail to update in hardware even though the parent policy does update. If this occurs, you can make additional changes to the policy and its applications to correct the discrepancy until the error messages are no longer displayed. Alternatively consider using command `checkpoint-rollback` as described in the AOS-CX Fundamentals Guide.

Policies using the `per-interface` parameter have slightly different warning and in-progress messages due to unique instances of the policy being created and applied to individual physical interfaces and LAGs.

For example, this is how the warning messages will appear if the unique instances of the policy for interfaces 1/1/2-1/1/3 fail to update while the unique instances of the policy for interfaces 1/1/1,1/1/4 successfully update.

```
switch(config)# show policy commands
! policy my_policy user configuration does not match active configuration
on interface 1/1/2 for ingress.
! policy my_policy user configuration does not match active configuration
on interface 1/1/3 for ingress.
policy my_policy
  10 class ip my_ip_class action drop
interface 1/1/1
  apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
interface 1/1/2
  apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
interface 1/1/3
  apply policy my_policy in per-interface
interface 1/1/4
  apply policy my_policy in per-interface
switch(config)# show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
% Warning: my_policy user configuration does not match active configuration
```

```

on interface 1/1/2 for ingress.
% Warning: my_policy user configuration does not match active configuration
on interface 1/1/3 for ingress.
    my_policy
    10
    my_ip_class ipv4
        drop

```

This is how the in-progress messages will appear if the child policies for interfaces 1/1/2-1/1/3 are currently updating while the child policies for interfaces 1/1/1,1/1/4 have successfully updated.

```

switch(config)# show policy commands
! policy my_policy user configuration currently being processed on
interface 1/1/2 for ingress.
! policy my_policy user configuration currently being processed on
interface 1/1/3 for ingress.
! run 'show policy [commands]' to display active policy configuration.
policy my_policy
    10 class ip my_ip_class action drop
interface 1/1/1
    apply policy my_policy in per-interface
! policy my_policy user configuration currently being processed
! run 'show policy [commands]' to display active policy configuration.
interface 1/1/2
    apply policy my_policy in per-interface
! policy my_policy user configuration currently being processed
! run 'show policy [commands]' to display active policy configuration.
interface 1/1/3
    apply policy my_policy in per-interface
interface 1/1/4
    apply policy my_policy in per-interface
switch(config)# show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
% Warning: my_policy user configuration currently being processed on
interface 1/1/2 for ingress.
% Warning: my_policy user configuration currently being processed on
interface 1/1/3 for ingress.
%      run 'show policy [commands]' to display active policy
configuration.

```

```

    my_policy
10
    my_ip_class ipv4
        drop

```

This is how the warning messages will appear if the child policies for interfaces 1/1/2-1/1/3 failed to apply or replace while the child policies for interfaces 1/1/1,1/1/4 have successfully applied or replaced.

```

switch(config)# show policy commands
policy my_policy
    10 class ip my_ip_class action drop
interface 1/1/1
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
interface 1/1/2
    apply policy my_policy in per-interface
! policy my_policy user configuration does not match active configuration.
! run 'policy NAME reset' to reset policy to match active configuration.
interface 1/1/3
    apply policy my_policy in per-interface
interface 1/1/4
    apply policy my_policy in per-interface
switch(config)# show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
      my_policy
10
      my_ip_class ipv4
          drop

```

Classifier policy application

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
L2 interface (port)	Yes		Yes	Yes	Yes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
L2 LAG	Yes		Yes	Yes	Yes
L3 interface (port)	Yes	Yes	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes	Yes	Yes
L3 interface (port) subinterface	Yes		Yes		
L3 LAG subinterface	Yes		Yes		
VLAN	Yes		Yes	Yes	Yes
Interface VLAN		Yes (PBR)			



NOTE

The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

PCP on MAC classes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
L2 interface (port)	Yes		Yes	Yes	Yes
L2 LAG	Yes		Yes	Yes	Yes
L3 interface (port)	Yes	Yes	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes	Yes	Yes
VLAN	Yes		Yes	Yes	Yes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
Interface VLAN		Yes (PBR)			



NOTE

The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

PCP on MAC classes

Classifier policies can be applied as follows:

Policy type Direction	IPv4 In	IPv6 In
L2 interface (port)	Yes	Yes
L2 LAG	Yes	Yes
VLAN	Yes	Yes



NOTE

Port policies and port-access client policies cannot be configured at the same time.

Classifier policies can be applied as follows:

Policy type Direction	IPv4 In	IPv6 In	MAC in
L2 interface (port)	Yes	Yes	Yes
L2 LAG	Yes	Yes	Yes
VLAN	Yes	Yes	Yes



NOTE

Port policies and port-access client policies cannot be configured at the same time.

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes
L3 interface (port) subin terface	Yes	Yes	Yes
L3 LAG subinterface	Yes	Yes	Yes
VLAN	Yes		Yes
Interface VLAN		Yes (PBR)	



NOTE

Policies cannot match multicast packets on the routed in direction.

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

HPE Aruba Networking 8320, 8325, 9300/9300S, 10000, 10040 Switch Series:

TCP flags CWR and ECE

TCP flags and TTL (hop limit) on IP classes

Fragment on IPv6 classes in VLAN policies

VLAN ID in VLAN policies

HPE Aruba Networking 8320 Switch Series only:

IPv4 AH on inbound classes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	MAC In
VLAN	Yes		Yes
Interface VLAN		Yes (PBR)	



NOTE

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

- TCP flags CWR and ECE
- TCP flags and TTL (hop limit) on IP classes
- Fragment on IPv6 classes in VLAN policies
- VLAN ID in VLAN policies

Classifier policy commands

Select a command from the list in the left navigation menu.

Subtopics

Classifier policy application

apply policy (config-if, config-lag-if, config-vlan)

apply policy

class copy

class ip

class ipv6

class mac

class resequence

class reset

clear policy hitcounts

policy

policy copy

policy resequence

policy reset

show class

show policy

Classifier policy application

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
L2 interface (port)	Yes		Yes	Yes	Yes
L2 LAG	Yes		Yes	Yes	Yes
L3 interface (port)	Yes	Yes	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes	Yes	Yes
L3 interface (port) subinterface	Yes		Yes		
L3 LAG subinterface	Yes		Yes		
VLAN	Yes		Yes	Yes	Yes
Interface VLAN		Yes (PBR)			



NOTE

The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

PCP on MAC classes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
L2 interface (port)	Yes		Yes	Yes	Yes
L2 LAG	Yes		Yes	Yes	Yes
L3 interface (port)	Yes	Yes	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes	Yes	Yes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	IPv4+6 Out	MAC In	MAC Out
VLAN	Yes		Yes	Yes	Yes
Interface VLAN	Yes (PBR)				



NOTE

The following match criteria is not supported. If this match criteria is attempted to be configured, an error message will be displayed and the action will not be completed.

PCP on MAC classes

Classifier policies can be applied as follows:

Policy type Direction	IPv4 In	IPv6 In
L2 interface (port)	Yes	Yes
L2 LAG	Yes	Yes
VLAN	Yes	Yes



NOTE

Port policies and port-access client policies cannot be configured at the same time.

Classifier policies can be applied as follows:

Policy type Direction	IPv4 In	IPv6 In	MAC in
L2 interface (port)	Yes	Yes	Yes
L2 LAG	Yes	Yes	Yes

Policy type Direction	IPv4 In	IPv6 In	MAC in
VLAN	Yes	Yes	Yes



NOTE

Port policies and port-access client policies cannot be configured at the same time.

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes
L3 interface (port) subin terface	Yes	Yes	Yes
L3 LAG subinterface	Yes	Yes	Yes
VLAN	Yes		Yes
Interface VLAN		Yes (PBR)	



NOTE

Policies cannot match multicast packets on the routed in direction.

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

HPE Aruba Networking 8320, 8325, 9300/9300S, 10000, 10040 Switch Series:

TCP flags CWR and ECE

TCP flags and TTL (hop limit) on IP classes

Fragment on IPv6 classes in VLAN policies

VLAN ID in VLAN policies

HPE Aruba Networking 8320 Switch Series only:

IPv4 AH on inbound classes

Policy type Direction	IPv4+6 In	IPv4+6 Rt - In	MAC In
L2 interface (port)	Yes		Yes
L2 LAG	Yes		Yes
L3 interface (port)	Yes	Yes	Yes
L3 LAG	Yes	Yes	Yes
VLAN	Yes		Yes
Interface VLAN		Yes (PBR)	



NOTE

The following match criteria are not supported. If any of these match criteria are attempted to be configured, an error message will be displayed and the action will not be completed.

TCP flags CWR and ECE

TCP flags and TTL (hop limit) on IP classes

Fragment on IPv6 classes in VLAN policies

VLAN ID in VLAN policies

apply policy (config-if, config-lag-if, config-vlan)

Syntax

Context config-if, config-lag-if:

```
apply policy <POLICY-NAME> {in} [per-interface]
no apply policy <POLICY-NAME> {in} [per-interface]
```

Context config-vlan:

```
apply policy <POLICY-NAME> in
no apply policy <POLICY-NAME> in
```

Description

Applies a policy to the current physical interface port or LAG or VLAN context.

The **no** form of this command removes a policy from the interface or VLAN specified by the current context.

Parameter	Description
<code><POLICY-NAME></code>	Specifies the policy to apply.
<code>in</code>	Selects the inbound (ingress) traffic direction.
<code>per-interface</code>	Specifies that unique instances of the policy be applied to each interface or LAG rather than the default of sharing the policy across all interfaces and LAGs.

Usage (applies to config-if, config-lag-if contexts)

- When **per-interface** is included, unique instances of the policy are applied to each physical interface port or LAG rather than the default of sharing the policy across all interfaces and LAGs. The unique instance of a policy has a parent-child relationship with the policy from which it was created. The **per-interface** option is useful when you want unique policers to be created for each interface or LAG rather than using shared policers. It is also useful when you want the statistics (hit counts and conform rate) to be specific to an interface or LAG rather than being aggregated. Because **per-interface** creates more hardware instances of a policy, resource consumption may increase significantly. It is recommended that you use **show resources** to monitor resource utilization as configuration is applied.

Usage (applies to config-vlan context)

- Only one policy type may be applied to a VLAN at a time. Therefore, using the **apply policy** command on a VLAN with an already-applied policy of the same type, will replace the applied policy.

Examples On the HPE Aruba Networking 6400 Switch Series, interface identification differs. Applying a policy to an interface (ingress):

```
switch(config) # interface 1/1/1
switch(config-if) # apply policy MY_POLICY1 in
```

Applying a policy to an interface (ingress) specifying **per-interface**:

```
switch(config) # interface 1/1/2
switch(config-if) # apply policy MY_POLICY1 in per-interface
```

Applying a policy to an interface (egress): Applying a policy to an interface (egress) specifying `per-interface`: Applying a policy to an interface range (ingress):

```
switch(config) # interface 1/1/3-1/1/6
switch(config-if-<1/1/2-1/1/5>) # apply policy MY_POLICY3 in
```

Applying a policy to an interface range (ingress) specifying **per-interface**:

```
switch(config) # interface 1/1/7-1/1/9  
switch(config-if-<1/1/2-1/1/5>) # apply policy MY_POLICY4 in per-interface
```

Removing a policy from an interface (ingress):

```
switch(config) # interface 1/1/1  
switch(config-if) # no apply policy MY_POLICY1 in
```

Removing a policy from an interface range (ingress):

```
switch(config) # interface 1/1/3-1/1/6  
switch(config-if-<1/1/3-1/1/6>) # no apply policy MY_POLICY3 in
```

Applying a policy to a subinterface (ingress): Applying a policy to a subinterface (egress): Applying a policy to a LAG (ingress):

```
switch(config) # interface lag 100  
switch(config-lag-if) # apply policy MY_POLICY5 in
```

Applying a policy to a LAG (ingress) specifying **per-interface**:

```
switch(config) # interface lag 200  
switch(config-lag-if) # apply policy MY_POLICY5 in per-interface
```

Removing a policy from a LAG (ingress):

```
switch(config) # interface lag 100  
switch(config-lag-if) # no apply policy MY_POLICY5 in
```

Applying a policy to a LAG subinterface (ingress): Applying a policy to a LAG subinterface (egress): Applying a policy to a VLAN (ingress):

```
switch(config) # vlan 1  
switch(config-vlan) # apply policy MY_POLICY6 in
```

Applying a policy to a VLAN (egress):

```
switch(config) # vlan 1  
switch(config-vlan) # no apply policy my_policy out
```

Applying a policy to multiple VLANs (ingress):

```
switch(config) # vlan 10,20  
switch(config-vlan-<10,20>) # apply policy MY_POLICY7 in
```

Applying a policy to an interface VLAN routed (ingress): Applying a policy to an interface VLAN range routed (ingress): Removing a policy from a VLAN (ingress):

```
switch(config) # vlan 1  
switch(config-vlan) # no apply policy MY_POLICY6 in
```

Removing a policy from multiple VLANs (ingress):

```
switch(config)# vlan 10,20  
switch(config-vlan-<10,20>)# no apply policy MY_POLICY7 in
```

Removing a policy from an interface VLAN routed (ingress): Removing a policy from an interface VLAN range routed (ingress):

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config-if config-lag-if config-vlan	Administrators or local user group members with execution rights for this command.

apply policy

Syntax

```
apply policy <POLICY-NAME> {in}  
no apply policy <POLICY-NAME> {in}
```

Description

Applies a policy to the global config context.

Only one policy can be globally applied at a time. Applying a policy globally again, replaces the previous globally applied policy.

The **no** form of this command removes application of the global policy.

Parameter	Description
<POLICY-NAME>	Specifies the policy to apply.

Parameter	Description
in	Selects the inbound (ingress) traffic direction.

Examples

Applying policy global1 to the global config context:

```
switch(config)# apply policy global1 in
```

Removing application of policy global1 from the global config context:

```
switch(config)# no apply policy global1 in
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i 6000 6100	config	Administrators or local user group members with execution rights for this command.

class copy

Syntax

```
class {ip|ipv6|mac} <CLASS-NAME> copy <DESTINATION-CLASS>
```

Description

Copies a class to a new destination class or overwrites an existing class. Copying a class copies all entries as well.

Parameter	Description
<pre> {ip ipv6 mac } <CLASS-NAME> </pre>	Specifies the type and name of the class to be copied.
<pre> <DESTINATION-CLASS> </pre>	Specifies the name of the destination class.

Examples

Copying an IPv4 class. Copying a class with entries copies all its entries as well:

```

switch(config)# class ip MY_IP_CLASS copy MY_IP_CLASS2
switch(config)# do show class
Type          Name
Sequence      Comment
              Action          L3 Protocol
              Source IP Address Source L4 Port(s)
              Destination IP Address or FQDN Destination L4 Port(s)
              Additional Parameters
-----
---
IPv4          MY_IP_CLASS
              11  ignore          udp
              any
              any
              21  match          tcp
              192.168.0.1
              192.168.0.2
-----
---
IPv4          MY_IP_CLASS2
              11  ignore          udp
              any

```

```

any
21 match tcp
192.168.0.1
192.168.0.2

```

Copying an IPv6 class:

```

switch(config)# class ipv6 MY_IPV6_CLASS copy MY_IPV6_CLASS2
switch(config)# do show class
Type          Name
Sequence Comment
Action          L3 Protocol
Source IP Address Source L4 Port(s)
Destination IP Address or FQDN Destination L4 Port(s)
Additional Parameters
-----
---
IPv6          MY_IPV6_CLASS
2 ignore      udp
any
any
-----
---
IPv6          MY_IPV6_CLASS2
2 ignore      udp
any
any

```

Copying a MAC class:

```

switch(config)# class mac MY_MAC_CLASS copy MY_MAC_CLASS2
switch(config)# do show class
Type          Name
Sequence Comment
Action          EtherType
Source MAC Address
Destination MAC Address
Additional Parameters
-----
---
MAC          MY_MAC_CLASS
2 ignore      arp
any
any
-----
---
MAC          MY_MAC_CLASS2

```

```
2 ignore
any
any
arp
```

Command History

Release	Modification
10.17.1000	Added support for FQDN.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

class ip

Syntax

Syntax to create an IPv4 class and enter its context. Plus syntax to remove a class:

```
class ip <CLASS-NAME>
no class ip <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols **ah**, **eigrp**, **esp**, **gre**, **igmp**, **l2tp**, **ospf**, **pim**, **vrrp** (**ip** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ah|eigrp|esp|gre|igmp|ip|l2tp|ospf|pim|vrrp|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]}
{any|<DST-IP-ADDRESS>[/{<PREFIX-LENGTH>|<SUBNET-MASK>}]|fqdn <fqdn>}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols **sctp**, **tcp**, **udp**:

```

[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |fqdn <fqdn>}
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
[cwr] [ece] [urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>

```

Syntax (within the class context) for creating or removing class entries for protocol **icmp**:

```

[<SEQUENCE-NUMBER>]
{match|ignore}
{icmp}
{any|<SRC-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] }
{any|<DST-IP-ADDRESS>[/ {<PREFIX-LENGTH>|<SUBNET-MASK>}] |fqdn <fqdn>}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-
VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>

```

Syntax (within the class context) for class entry comments:

```

[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment

```

Description

Creates or modifies an IPv4 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, the class can classify traffic based on IPv4 header information.

The **no** form of the command can be used to delete either an IPv4 traffic class (use **no** with the class command) or an individual IPv4 traffic class entry (use **no** with the sequence number).

Parameter	Description
ip	Specifies create or modify an IPv4 class.

Parameter	Description
<code><CLASS-NAME></code>	Specifies the name of this class.
<code><SEQUENCE-NUMBER></code>	Specifies a sequence number for the class entry. Optional. Range: 1 - 4294967295.
<code>{match ignore}</code>	Creates a rule to match or ignore specified packets.
<code><IP-PROTOCOL-NUM></code>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
<code>{any <SRC-IP-ADDRESS></code> <code>[/ {<PREFIX-LENGTH> <SUBNET-MASK>}] }</code>	Specifies the source IPv4 address. • any - specifies any source IPv4 address. • <SRC - IP - ADDRESS> - specifies the source IPv4 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<code>{any <DST-IP-ADDRESS></code> <code>[/ {<PREFIX-LENGTH> <SUBNET-MASK>}] }</code>	Specifies the destination IPv4 address, network address, fqdn or any. • any - specifies any destination IPv4 address. • <DST - IP - ADDRESS> - specifies the destination IPv4 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<code>[{eq gt lt} <PORT> range</code> <code><MIN-PORT></code> <code><MAX-PORT>]</code>	Specifies the port or port range. Port numbers are in the range of 0 to 65535. • eq <PORT> - specifies the Layer 4 port. • gt <PORT> - specifies any Layer 4 port greater than the indicated port. • lt <PORT> - specifies any Layer 4 port less than the indicated port. • range <MIN - PORT> <MAX - PORT> - specifies the Layer 4 port range.

Parameter	Description
cwr	Specifies matching on the TCP Flag CWR : Congestion Window Reduced
ece	Specifies matching on the TCP Flag ECE : Explicit Congestion Notification [ECN] - Echo
urg	Specifies matching on the TCP Flag: Urgent.
ack	Specifies matching on the TCP Flag: Acknowledgment.
psh	Specifies matching on the TCP Flag: Push buffered data to receiving application.
rst	Specifies matching on the TCP Flag: Reset the connection.
syn	Specifies matching on the TCP Flag: Synchronize sequence numbers.
fin	Specifies matching on the TCP Flag: Finish connection.
established	Specifies matching on the TCP Flag: Established connection.
dscp <DSCP-SPECIFIER>	Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP-VALUE> (0 to 63) or one of these keywords: • AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability) • AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability) • AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability) • AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability) • AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability) • AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability) • AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability) • AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability) • AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability)

Parameter	Description
	• AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability) • AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability) • CS0 - DSCP 0 (Class Selector 0: Default) • CS1 - DSCP 8 (Class Selector 1: Scavenger) • CS2 - DSCP 16 (Class Selector 2: OAM) • CS3 - DSCP 24 (Class Selector 3: Signaling) • CS4 - DSCP 32 (Class Selector 4: Real time) • CS5 - DSCP 40 (Class Selector 5: Broadcast video) • CS6 - DSCP 48 (Class Selector 6: Network control) • CS7 - DSCP 56 (Class Selector 7) • EF - DSCP 46 (Expedited Forwarding)
<code>ip-precedence <IP-PRECEDENCE-VALUE></code>	Specifies an IP precedence value. Range: 0 to 7.
<code>tos <TOS-VALUE></code>	Specifies the Type of Service value. Range: 0 to 31.
<code>fragment</code>	Specifies a fragment packet.
<code>vlan <VLAN-ID></code>	Specifies VLAN tag to match on. 802.1Q VLAN ID.
 NOTE This parameter cannot be used in any class that will be applied to a VLAN.	
<code>count</code>	Keeps the hit counts of the number of packets matching this class entry.
<code>[<SEQUENCE-NUMBER>] comment <TEXT-STRING></code>	Adds a comment to a class entry. The no form removes only the comment from the class entry.

Usage

- Entering an existing **<CLASS-NAME>** value will cause the existing class to be modified, with any new **<SEQUENCE-NUMBER>** value creating an additional class entry, and any existing **<SEQUENCE-NUMBER>** value replacing the existing class entry with the same sequence number.
- If no sequence number is specified, a new class entry will be appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the **<IP-PROTOCOL-NUM>** parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv4 class with invalid fqdn:

```
switch(config)# class ip my_ip_class
switch(config-class-ip)# match any any fqdn-group abc
Invalid destination FQDN address.
```

Creating an IPv4 class with three entries:

```
switch(config)# class ip my_ip_class
switch(config-class-ip)# 1 match icmp any any
switch(config-class-ip)# 2 ignore udp any any
switch(config-class-ip)# 3 match tcp 192.168.0.1 192.168.0.2
switch(config-class-ip)# exit
switch(config)# show class ip my_ip_class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address or FQDN	Destination L4 Port(s)	Additional Parameters

IPv4	my_ip_class									
		1	match		icmp					
			any							
			any							
		2	ignore		udp					
			any							
			any							
		3	match		tcp					
						192.168.0.1				
						192.168.0.2				

Adding a comment to an existing IPv4 class entry:

```

switch(config)# class ip my_ip_class
switch(config-class-ip)# 3 comment myipClass
switch(config-class-ip)# exit
switch(config)# show class ip my_ip_class

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address or FQDN	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	--------------------------------	------------------------	-----------------------

```

-----
---
IPv4      my_ip_class
          1 match                                icmp
            any
            any
          2 ignore                                udp
            any
            any
          3 myipClass
            match                                tcp
            192.168.0.1
            192.168.0.2

```

Removing a comment from an existing IPv4 class entry:

```

switch(config)# class ip my_ip_class
switch(config-class-ip)# no 3 comment
switch(config-class-ip)# exit
switch(config)# show class ip my_ip_class

```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address or FQDN	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	--------------------------------	------------------------	-----------------------

```

-----
---
IPv4      my_ip_class
          1 match                                icmp
            any
            any
          2 ignore                                udp
            any
            any

```

```

3 match                                     tcp
192.168.0.1
192.168.0.2
Type      Name
Sequence Comment
Action                                          L3 Protocol
Source IP Address                          Source L4 Port(s)
Destination IP Address                    Destination L4 Port(s)
Additional Parameters
-----
---
IPv4      my_ip_class
1 match                                     icmp
any
any
2 ignore                                     udp
any
any
3 match                                     tcp
192.168.0.1
192.168.0.2

```

Replacing an IPv4 class entry in an existing class:

```

switch(config)# class ip my_ip_class
switch(config-class-ip)# 10 match igmp any any
switch(config-class-ip)# exit
switch(config)# show class ip my_ip_class
Type      Name
Sequence Comment
Action                                          L3 Protocol
Source IP Address                          Source L4 Port(s)
Destination IP Address or FQDN            Destination L4 Port(s)
Additional Parameters
-----
---
IPv4      my_ip_class
10 match                                     igmp
any
any
20 ignore                                     udp
any
any
30 match                                     tcp

```

192.168.0.1

192.168.0.2

Removing an IPv4 class entry:

```
switch(config)# class ip MY_IP_CLASS
switch(config-class-ip)# no 10
switch(config-class-ip)# exit
switch(config)# show class ip my_ip_class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address or FQDN	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	--------------------------------	------------------------	-----------------------

IPv4	my_ip_class									
		20	ignore		udp					
			any							
			any							
		30	match		tcp					
						192.168.0.1				
						192.168.0.2				

Removing an IPv4 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE

The corresponding entries are only removed if the class is unused by all policy entries.

Creating an IPv4 class with FQDN:

```
switch(config)# class ip my_ip_class
switch(config-class-ip)# 1 match icmp any fqdn www.example.com
switch(config-class-ip)# exit
switch(config-class-ip)# show class ip my_ip_class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address or FQDN	Destination L4 Port(s)	Additional Entry Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	--------------------------------	------------------------	-----------------------------

```

---
IPv4      my_ip_class
          1  match                                icmp
              any
              www.example.com

```

Modifying an existing class to include FQDN:

```

switch(config)# class ip my_ip_class
switch(config-class-ip)# 2 match udp any fqdn www.anotherexample.com
switch(config-class-ip)# exit
switch(config-class-ip)# show class ip my_ip_class
Type      Name
Sequence Comment
          Action                                L3 Protocol
          Source IP Address                    Source L4 Port(s)
          Destination IP Address or FQDN        Destination L4
Port(s)
          Additional Parameters
-----
---
IPv4      my_ip_class
          1  match                                icmp
              any
              www.example.com
          2  match                                udp
              any
              www.anotherexample.com

```

Creating an IPv4 class with a long FQDN:

```

switch(config)# class ip my_long_fqdn_class
switch(config-class-ip)# 1 match icmp any fqdn
www.longsubdomainnamefornetworkconfigurationandtestingpurposes
withadditionaldetailsincludedforexampleusage.com
switch(config-class-ip)# exit
switch(config-class-ip)# show class ip my_long_fqdn_class
Type      Name
Sequence Comment
          Action                                L3 Protocol
          Source IP Address                    Source L4 Port(s)
          Destination IP Address or FQDN        Destination L4
Port(s)
          Additional Entry Parameters
-----
---
IPv4      my_long_fqdn_class

```

```

1 match icmp
any
www.longsubdomainnamefornetworkconfigurationandtestingpur...

```

Creating an IPv6 class FQDN:

```

switch(config)# class ipv6 my_ipv6_class
switch(config-class-ip)# 1 match icmpv6 any fqdn www.example.com
switch(config-class-ip)# exit
switch(config-class-ip)# show class ipv6 my_ipv6_class

```

Type	Name
Sequence	Comment
Action	L3 Protocol
Source IP Address	Source L4 Port(s)
Destination IP Address or FQDN	Destination L4
Port(s)	
Additional Entry Parameters	

```

-----
---
IPv6      my_ipv6_class
1 match          icmpv6
any
www.example.com

```

Modifying an existing class to include FQDN (IPv6):

```

switch(config)# class ipv6 my_ipv6_class
switch(config-class-ip)# 2 match any any fqdn www.anotherexample.com
switch(config-class-ip)# exit
switch(config-class-ip)# show class ipv6 my_ipv6_class

```

Type	Name
Sequence	Comment
Action	L3 Protocol
Source IP Address	Source L4 Port(s)
Destination IP Address or FQDN	Destination L4
Port(s)	
Additional Parameters	

```

-----
---
IPv6      my_ipv6_class
1 match          icmpv6
any
www.example.com
2 match          any
any
www.anotherexample.com

```

Command History

Release	Modification
10.17.1000	Added support for FQDN. Added support for EIGRP, L2TP and VRRP protocols.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> The class ip <CLASS - NAME> command takes you into the config - class - ipconfig - class - ip context where you enter the class entries.	Administrators or local user group members with execution rights for this command.

class ipv6

Syntax

Syntax to create an IPv6 class and enter its context. Plus syntax to remove a class:

```
class ipv6 <CLASS-NAME>
no class ipv6 <CLASS-NAME>
```

Syntax (within the class context) for creating or removing class entries for protocols **ah**, **eigrp**, **esp**, **gre**, **l2tp**, **ospf**, **pim**, **vrrp**, (**ipv6** is available as an alias for **any**):

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{any|ah|eigrp|esp|gre|ipv6|l2tp|ospf|pim|vrrp|<IP-PROTOCOL-NUM>}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocols **sctp**, **tcp**, **udp**:

```
[<SEQUENCE-NUMBER>]
{match|ignore}
{sctp|tcp|udp}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
[{eq|gt|lt} <PORT>|range <MIN-PORT>
    <MAX-PORT>]
[urg] [ack] [psh] [rst] [syn] [fin] [established]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for creating or removing class entries for protocol **icmpv6**:

```
[<SEQUENCE-NUMBER>]
{permit|deny}
{icmpv6}
{any|<SRC-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
{any|<DST-IP-ADDRESS>[/<PREFIX-LENGTH>|<SUBNET-MASK>]}
[icmp-type {echo|echo-reply|<ICMP-TYPE-VALUE>}] [icmp-code <ICMP-CODE-VALUE>]
[dscp <DSCP-SPECIFIER>] [ip-precedence <IP-PRECEDENCE-VALUE>]
[tos <TOS-VALUE>] [fragment] [vlan <VLAN-ID>] [count]
no <SEQUENCE-NUMBER>
```

Syntax (within the class context) for class entry comments:

```
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>
no <SEQUENCE-NUMBER> comment
```

Description

Creates or modifies an IPv6 traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on IPv6 header information.

The **no** form of the command deletes either an IPv6 traffic class (use **no** with the class command) or an individual IPv6 traffic class entry (use **no** with the sequence number).

Parameter	Description
ipv6	Specifies create or modify an IPv6 class.

Parameter	Description
<code><CLASS-NAME></code>	Specifies the name of this class.
<code><SEQUENCE-NUMBER></code>	Specifies a sequence number for the class entry. Optional. Range: 1 - 4294967295.
<code>{match ignore}</code>	Creates a rule to match or ignore specified packets.
<code><IP-PROTOCOL-NUM></code>	Specifies the protocol as its Internet Protocol number. For example, 2 corresponds to the IGMP protocol. Range: 0 to 255.
<code>{any <SRC-IP-ADDRESS>[/{<PREFIX-LENGTH> <SUBNET-MASK>}]}</code>	Specifies the source IPv6 address. • any - specifies any source IPv6 address. • <SRC - IP - ADDRESS> - specifies the source IPv4 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<code>{any <DST-IP-ADDRESS>[/{<PREFIX-LENGTH> <SUBNET-MASK>}]}</code>	Specifies the destination IPv4 address. • any - specifies any destination IPv6 address. • <DST - IP - ADDRESS> - specifies the destination IPv6 host address. ◦ <PREFIX - LENGTH> - specifies the address bits to mask (CIDR subnet mask notation). Range: 1 to 32. ◦ <SUBNET - MASK> - specifies the address bits to mask (dotted decimal notation).
<code>[{eq gt lt} <PORT> range <MIN-PORT><MAX-PORT>]</code>	Specifies the port or port range. Port numbers are in the range of 0 to 65535. • eq <PORT> - specifies the Layer 4 port. • gt <PORT> - specifies any Layer 4 port greater than the indicated port. • lt <PORT> - specifies any Layer 4 port less than the indicated port. • range <MIN - PORT> <MAX - PORT> - specifies the Layer 4 port range.
<code>cwr</code>	Specifies matching on the TCP Flag CWR : Congestion Window Reduced

Parameter	Description
ece	Specifies matching on the TCP Flag ECE : Explicit Congestion Notification [ECN] - Echo
urg	Specifies matching on the TCP Flag: Urgent.
ack	Specifies matching on the TCP Flag: Acknowledgment.
psh	Specifies matching on the TCP Flag: Push buffered data to receiving application.
rst	Specifies matching on the TCP Flag: Reset the connection.
syn	Specifies matching on the TCP Flag: Synchronize sequence numbers.
fin	Specifies matching on the TCP Flag: Finish connection.
established	Specifies matching on the TCP Flag: Established connection.
dscp <DSCP-SPECIFIER>	Specifies the Differentiated Services Code Point (DSCP), either a numeric <DSCP-VALUE> (0 to 63) or one of these keywords: • AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability) • AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability) • AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability) • AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability) • AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability) • AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability) • AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability) • AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability) • AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability) • AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability)

Parameter	Description
	• AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability) • CS0 - DSCP 0 (Class Selector 0: Default) • CS1 - DSCP 8 (Class Selector 1: Scavenger) • CS2 - DSCP 16 (Class Selector 2: OAM) • CS3 - DSCP 24 (Class Selector 3: Signaling) • CS4 - DSCP 32 (Class Selector 4: Real time) • CS5 - DSCP 40 (Class Selector 5: Broadcast video) • CS6 - DSCP 48 (Class Selector 6: Network control) • CS7 - DSCP 56 (Class Selector 7) • EF - DSCP 46 (Expedited Forwarding)
<code>ip-precedence <IP-PRECEDENCE-VALUE></code>	Specifies an IP precedence value. Range: 0 to 7.
<code>tos <TOS-VALUE></code>	Specifies the Type of Service value. Range: 0 to 31.
<code>fragment</code>	Specifies a fragment packet.
<code>vlan <VLAN-ID></code>	Specifies VLAN tag to match on. 802.1Q VLAN ID.
 NOTE This parameter cannot be used in any class that will be applied to a VLAN.	
<code>count</code>	Keeps the hit counts of the number of packets matching this class entry.
<code>[<SEQUENCE-NUMBER>]</code> <code>comment <TEXT-STRING></code>	Adds a comment to a class entry. The no form removes only the comment from the class entry.

Usage

- If you enter an existing **<CLASS-NAME>** value, the existing class is modified with any new **<SEQUENCE-NUMBER>** value. This action creates an additional class entry. Any existing **<SEQUENCE-NUMBER>** value replaces the existing class entry with the same sequence number.

- If no sequence number is specified, a new class entry is appended to the end of the class with a sequence number equal to the highest class entry currently in the list plus 10.
- If the **<IP-PROTOCOL-NUM>** parameter is used instead of a protocol name, ensure that any needed class entry-definition parameters specific to the selected protocol are also provided.

Examples

Creating an IPv6 class with two entries:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv6	MY_IPV6_CLASS	10	match		icmpv6	any		any		
		20	ignore		udp	any		any		

Adding a comment to an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match icmpv6 any any
switch(config-class-ipv6)# 20 ignore udp any any
switch(config-class-ipv6)# 20 comment myipv6class
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters


```
IPv6      MY_IPV6_CLASS
10 match
    any
    any
20 myipv6class
    ignore
    any
    any
```

Removing a comment from an existing IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 20 comment
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------


```
IPv6      MY_IPV6_CLASS
10 match
    any
    any
20 ignore
    any
    any
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters
------	------	----------	---------	--------	-------------	-------------------	-------------------	------------------------	------------------------	-----------------------


```
IPv6      MY_IPV6_CLASS
10 match
    any
    any
20 ignore
    any
    any
```

Replacing an IPv6 class entry in an existing IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# 10 match any any 1020::
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv6	MY_IPV6_CLASS									
		10	match				any			
				any						
				1020::						
		20	ignore				udp			
				any						
				any						

Removing an IPv6 class entry:

```
switch(config)# class ipv6 MY_IPV6_CLASS
switch(config-class-ipv6)# no 10
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name	Sequence	Comment	Action	L3 Protocol	Source IP Address	Source L4 Port(s)	Destination IP Address	Destination L4 Port(s)	Additional Parameters

IPv6	MY_IPV6_CLASS									
		20	ignore				udp			
				any						
				any						

Removing an IPv6 class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE

The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class ipv6 MY_IPV6_CLASS
switch(config)# do show class
No Class found.
```

Command History

Release	Modification
10.17.1000	Added support for EIGRP, L2TP and VRRP protocols.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config The class ipv6 <CLASS - NAME> command takes you into the config - class - ipv6 command context where you enter the class entries.	Administrators or local user group members with execution rights for this command.

class mac

Syntax

```
class mac <CLASS-NAME>
  [<SEQUENCE-NUMBER>]
  {match|ignore}
  {any}
  {any}
```

```

{any|arp|appletalk|arp|fcoe|fcoe-init|ip|ipv6|ipx-arpa|ipx-non-arpa|is-
is|
    lldp|mpls-multicast|mpls-unicast|q-in-q|rbridge|trill|wake-on-lan|
    <NUMERIC-ETHERTYPE>}
[pcp <PCP-VALUE>] [vlan <VLAN-ID>] [count]
[<SEQUENCE-NUMBER>] comment <TEXT-STRING>

```

Description

Creates or modifies a MAC traffic class to match specified packets. Class is composed of one or more class entries ordered and prioritized by sequence numbers. With this command, each class can classify traffic based on MAC header information.

The **no** form of the command can be used to delete either a MAC traffic class (use **no** with the class command) or an individual MAC traffic class entry (use **no** with the sequence number).

Parameter	Description
mac	Specifies create or modify a MAC class.
<CLASS-NAME>	Specifies the name of this class.
<SEQUENCE-NUMBER>	Specifies a sequence number for the class entry. Optional. Range: 1 - 4294967295.
{match ignore}	Creates a rule to match or ignore specified packets.
comment	Stores the remaining entered text as a class comment.
{any}	Specifies any source MAC.
{any}	Specifies any destination MAC.
Protocol	Select an ethertype protocol from the following (enter one only): any - Any ethertype protocol <NUMERIC-ETHERTYPE> - Enter an EtherType protocol number. Range: 0x600 - 0xffff.
count	Keeps the hit counts of the number of packets matching this class entry.

Examples

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# match any any lldp
switch(config-class-mac)# ignore any any arp
switch(config-class-mac)# exit
switch(config)# do show class
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
---
MAC           MY_MAC_CLASS
              10 match                      lldp
                  any
                  any
              20 ignore                      arp
                  any
                  any
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 comment MY_CLASS_ENTRY10 comment MY_CLASS_ENTRY
switch(config-class-mac)# exit
switch(config)# do show class
Type          Name
Sequence      Comment
              Action                      EtherType
              Source MAC Address
              Destination MAC Address
              Additional Parameters
-----
---
MAC           MY_MAC_CLASS
              10 MY_CLASS_ENTRY
                  match                      lldp
                  any
                  any
              20 ignore                      arp
                  any
                  any
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 10 comment MY_CLASS_ENTRY
```

```
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name			
Sequence	Comment	Action		EtherType
		Source MAC Address		
		Destination MAC Address		
		Additional Parameters		

```
-----
---
```

MAC	MY_MAC_CLASS			
10	match			lldp
	any			
	any			
20	ignore			arp
	any			
	any			

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# 10 match any any any
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name			
Sequence	Comment	Action		EtherType
		Source MAC Address		
		Destination MAC Address		
		Additional Parameters		

```
-----
---
```

MAC	MY_MAC_CLASS			
10	match			any
	any			
	any			
20	ignore			arp
	any			
	any			

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# no 1
switch(config-class-mac)# exit
switch(config)# do show class
```

Type	Name			
Sequence	Comment	Action		EtherType

	Source MAC Address	
	Destination MAC Address	
	Additional Parameters	

MAC	MY_MAC_CLASS	
	2 ignore	arp
	any	
	any	

Creating a MAC class with the source MAC address and destination MAC address as 'any'.

```
switch(config)# class mac MY_MAC_CLASS
switch(config-class-mac)# match any any
```

Removing a MAC class. Removing a class with entries removes all its entries as well. If a class associated with a policy entry (or multiple policy entries) is removed, the corresponding entries are also removed.



NOTE

The corresponding entries are only removed if the class is unused by all policy entries.

```
switch(config)# no class mac MY_MAC_CLASS
switch(config)# do show class
No Class found.
```

Command History

Release	Modification
10.12	Command introduced on the 4100i Switch series.

Command Information

Platforms	Command context	Authority
4100i	config The class mac <CLASS - NAME> command takes you into the config - class - mac context where you enter the class entries.	Administrators or local user group members with execution rights for this command.

class resequence

Syntax

```
class {ip|ipv6} <CLASS-NAME> resequence <STARTING-SEQUENCE-NUMBER>
                                     <INCREMENT>
```

Description

Resequencing numbering in an IPv4, or IPv6 class.

Parameter	Description
<pre>{ip ipv6 mac } <CLASS-NAME></pre>	Specifies the class where you want to resequence class entries.
<pre><STARTING-SEQUENCE-NUMBER></pre>	Specifies the sequence number to start resequencing from.
<pre><INCREMENT></pre>	Specifies how much to increment the sequence numbers by.

Examples

Resequencing an IPv4 class:

```
switch(config)# class ip MY_IP_CLASS resequence 1 10
switch(config)# do show class
Type           Name
Sequence      Comment
              Action
              L3 Protocol
```

	Source IP Address	Source L4 Port(s)
	Destination IP Address	Destination L4 Port(s)
	Additional Parameters	

IPv4	MY_IP_CLASS	
1	match	igmp
	any	
	any	
11	ignore	udp
	any	
	any	
21	match	tcp
	192.168.0.1	
	192.168.0.2	

Resequencing an IPv6 class:

```
switch(config)# class ipv6 MY_IPV6_CLASS resequence 1 1
switch(config-class-ipv6)# exit
switch(config)# do show class
```

Type	Name		
	Sequence	Comment	
	Action		L3 Protocol
	Source IP Address		Source L4 Port(s)
	Destination IP Address		Destination L4 Port(s)
	Additional Parameters		

IPv6	MY_IPV6_CLASS		
1	match		any
	any		
	1020::		
2	ignore		udp
	any		
	any		

Resequencing a MAC class:

```
switch(config)# class mac MY_MAC_CLASS resequence 1 1
switch(config)# do show class
```

Type	Name		
	Sequence	Comment	
	Action		EtherType
	Source MAC Address		
	Destination MAC Address		
	Additional Parameters		

```
-----
---
MAC          MY_MAC_CLASS
             1 match                      any
               any
               any
             2 ignore                      arp
               any
               any
```

Command History

Release	Modification
10.12	Support for the mac parameter introduced for the 4100i Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

class reset

Syntax

```
class { all | ip <CLASS-NAME> | ipv6 <CLASS-NAME>
      |mac <CLASS-NAME>
      } reset
```

Description

Changes the user-specified class configuration to match the active class configuration. Use this command when there is a discrepancy between what the user configured and what is active and accepted by the system.

Parameter	Description
<pre>{ all ip <CLASS-NAME> ipv6 <CLASS-NAME> mac <CLASS-NAME> }</pre>	Specifies either all classes be reset or specifies the type (ip for IPv4, or ipv6 for IPv6 or mac for MAC ACL) and name of the class to be reset.

Examples

Resetting the user-specified configuration to the active configuration:

```
switch(config)# class all reset
```

Command History

Release	Modification
10.12	Support for the mac parameter introduced for the 4100i Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

clear policy hitcounts

Syntax

Description

Clears the policy hit count statistics.

Parameter	Description
<code>all</code>	Selects all policies.
<code><POLICY-NAME></code>	Specifies the policy name.
<code>interface <IF-NAME></code>	Specifies the interface name.
<code>vlan <VLAN-ID></code>	Specifies the VLAN.
<code>in</code>	Specifies the inbound (ingress) traffic direction.
<code>global</code>	Selects the globally applied policy.

Examples

```
switch#
```

```
clear policy hitcounts my_policy int 1/1/1 in
```

```
switch#
```

```
show policy hitcounts my_policy
```

Statistics for Policy my_policy: Interface 1/1/1* (in): Hit Count Configuration 10 class ipv6 my_class1 action dscp af21 action drop 0 10 match any any any count * policy statistics are shared among each context type (interface, VLAN). For routed ingress, they are only shared within the same VRF. Use 'policy NAME copy' to create a new policy for separate statistics.

```
switch#
```

```
clear policy hitcounts my_policy int 1/1/1 in
```

```
switch#
```

```
show policy hitcounts my_policy
```

Statistics for Policy my_policy: Interface 1/1/1* (in): Hit Count Configuration 10 class ipv6 my_class1 action dscp af21 action drop 0 10 match any any any count * policy statistics are shared among each context type (interface, VLAN). Use 'policy NAME copy' to create a new policy for separate statistics.

```
switch#
```

```
clear policy hitcounts global
```

```
switch#
```

```
show policy hitcounts global
```

Statistics for Policy global1: Global Policy: Hit Count Configuration 10 class ipv6 my_class1 action mirror 0 10 match any any any count * policy statistics are shared among each context type (interface, VLAN). For routed ingress, they are only shared within the same VRF. Use 'policy NAME copy' to create a new policy for separate statistics.

switch#

clear policy hitcounts global

switch#

show policy hitcounts global

Statistics for Policy global1: Global Policy: Hit Count Configuration 10 class ipv6 my_class1 action mirror 0 10 match any any any count * policy statistics are shared among each context type (interface, VLAN). For routed ingress, they are only shared within the same VRF. Use 'policy NAME copy' to create a new policy for separate statistics.

switch#

clear policy hitcounts global

switch#

show policy hitcounts global

Statistics for Policy global1: Global Policy: Hit Count Configuration 10 class ipv6 my_class1 action mirror 0 10 match any any any count * policy statistics are shared among each context type (interface, VLAN). Use 'policy NAME copy' to create a new policy for separate statistics.

Clearing hit counts for policy **MY_IPv6_Policy** applied to VLAN 10 (ingress):

```
switch# clear policy hitcounts My_IPv6_Policy vlan 10 in
```

Clearing hit counts for all policies:

```
switch# clear policy hitcounts all
```

Command History

Release

Modification

10.07 or earlier

- -

Command Information

Platforms

Command context

Authority

All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
---------------	---------------------------------	--

policy

Syntax

```
policy <POLICY-NAME>
  [<SEQUENCE-NUMBER>]
  class {ip|ipv6|mac} <CLASS-NAME>
    action {<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}
    [{<REMARK-ACTIONS> | <POLICE-ACTIONS> | <OTHER-ACTIONS>}]
  [<SEQUENCE-NUMBER>]
  comment ...
```

Description

Creates or modifies classifier policy and policy entries. A policy is made up of one or more policy entries ordered and prioritized by sequence numbers. Each entry has an IPv4/IPv6/MAC class and zero or more policy actions associated with it.

A policy must be applied using the **apply** command.

The **no** form of the command can be used to delete either a policy (use **no** with the policy command) or an individual policy entry (use **no** with the sequence number).

Parameter	Description
<code><POLICY-NAME></code>	Specifies the name of the policy.
<code><SEQUENCE-NUMBER></code>	Specifies a sequence number for the policy entry. Optional. Range: 1 to 4294967295.
<code>comment</code>	Stores the remaining entered text as a policy entry comment.
<code>class {ip ipv6 mac} <CLASS-NAME></code>	Specifies a type of class, ip for IPv4, ipv6 for IPv6 and mac for a MAC policy. And specifies a class name.
<code><REMARK-ACTIONS></code>	Remark actions can be any of the following options: { cos <COS - VALUE> ip - precedence <IP - PRECEDENCE_VALUE> dscp <DSCP - VALUE> } where:
<code>cos <COS-VALUE></code>	Specifies the Class of Service (CoS) value.

Parameter	Description
<code>ip-precedence <IP-PRECEDENCE-VALUE></code>	Specifies the numeric IP precedence value. Range: 0 to 7.
<code>dscp <DSCP-VALUE></code>	Specifies a Differentiated Services Code Point (DSCP) value. Enter either a numeric value (0 to 63) or a keyword as follows: • AF11 - DSCP 10 (Assured Forwarding Class 1, low drop probability) • AF12 - DSCP 12 (Assured Forwarding Class 1, medium drop probability) • AF13 - DSCP 14 (Assured Forwarding Class 1, high drop probability) • AF21 - DSCP 18 (Assured Forwarding Class 2, low drop probability) • AF22 - DSCP 20 (Assured Forwarding Class 2, medium drop probability) • AF23 - DSCP 22 (Assured Forwarding Class 2, high drop probability) • AF31 - DSCP 26 (Assured Forwarding Class 3, low drop probability) • AF32 - DSCP 28 (Assured Forwarding Class 3, medium drop probability) • AF33 - DSCP 30 (Assured Forwarding Class 3, high drop probability) • AF41 - DSCP 34 (Assured Forwarding Class 4, low drop probability) • AF42 - DSCP 36 (Assured Forwarding Class 4, medium drop probability) • AF43 - DSCP 38 (Assured Forwarding Class 4, high drop probability) • CS0 - DSCP 0 (Class Selector 0: Default) • CS1 - DSCP 8 (Class Selector 1: Scavenger) • CS2 - DSCP 16 (Class Selector 2: OAM) • CS3 - DSCP 24 (Class Selector 3: Signaling) • CS4 - DSCP 32 (Class Selector 4: Real time) • CS5 - DSCP 40 (Class Selector 5: Broadcast video) • CS6 - DSCP 48 (Class Selector 6: Network control) • CS7 - DSCP 56 (Class Selector 7) • EF - DSCP 46 (Expedited Forwarding)

Parameter	Description
<code><POLICE-ACTIONS></code>	Police actions can be the following {cir <RATE - BPS> exceed} where:
<code>cir kbps <RATE-KBPS></code>	Specifies a Committed Information Rate value in Kilobits per second. Range: 1 to 4294967295.
	Specifies a Committed Burst Size value in bytes. Range: 1 to 4294967295.
<code><OTHER-ACTIONS></code>	Other actions can be the following:
<code>drop</code>	Specifies drop traffic.

Usage

- An applied policy will process a packet sequentially against policy entries in the list until the last policy entry in the list has been evaluated or the packet matches an entry.
- Entering an existing `<POLICY-NAME>` value will cause the existing policy to be modified, with any new `<SEQUENCE-NUMBER>` value creating an additional policy entry, and any existing `<SEQUENCE-NUMBER>` value replacing the existing policy entry with the same sequence number.
- If no sequence number is specified, a new policy entry will be appended to the end of the entry list with a sequence number equal to the highest policy entry currently in the list plus 10.

Examples

Creating a policy with several entries:

```
switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ipv6 MY_CLASS1 action dscp af21 action drop
switch(config-policy)# 20 class ip MY_CLASS3 action mirror 1
switch(config-policy)# exit
switch(config)# show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
```

```

MY_POLICY
10
    MY_CLASS1 ipv6
        drop
        dscp AF21
20
    MY_CLASS3 ipv4
        mirror 1

```

Adding a comment to an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 20 comment MY_TEST_POLICY
switch(config-policy)# exit
switch(config)# show policy

```

Name	Sequence	Comment	Class	Type	action

MY_POLICY	10		MY_CLASS1	ipv6	drop dscp AF21
	20	MY_TEST_POLICY	MY_CLASS3	ipv4	mirror 1

Removing a comment from an existing policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 20 comment
switch(config-policy)# exit
switch(config)# show policy

```

Name	Sequence	Comment	Class	Type	action

MY_POLICY	10		MY_CLASS1	ipv6	drop dscp AF21
	20		MY_CLASS3	ipv4	mirror 1

Adding/Replacing a policy entry in an existing policy:

```

switch(config)# policy MY_POLICY
switch(config-policy)# 10 class ip MY_CLASS3 action drop action dscp af21
switch(config-policy)# exit
switch(config)# show policy

```

Name	Sequence	Comment	Class	Type	action

MY_POLICY	10		MY_CLASS3	ipv4	

```

                drop
                dscp AF21
20
    MY_CLASS3 ipv4
        mirror 1

```

Removing a policy entry:

```

switch(config)# policy MY_POLICY
switch(config-policy)# no 10
switch(config-policy)# exit
switch(config)# show policy
    Name
Sequence Comment
    Class Type
        action
-----
---
    MY_POLICY
20
    MY_CLASS3 ipv4
        mirror 1

```

Removing a policy:

```

switch(config)# no policy MY_POLICY
switch(config)# show policy
Name          Sequence    Comment      Class      Type      action
-----
---
MY_POLICY     22                      MY_CLASS3  ipv4      mirror 1

```

The policer **exceed** DSCP action cannot be combined with other actions in the same policy entry, but other entries in the policy may use other actions.

For example, this configuration is valid:

```

switch(config)# policy my_policy
switch(config-policy)# 10 class ip my_class action cir kbps 1000 cbs 15625
exceed dscp EF

```

But this is not because it adds a secondary action within the same policy entry:

```

6300(config-policy)# 10 class ip my_class action cir kbps 1000 cbs 15625
exceed dscp EF action mirror 1

```

Invalid input: action

Command History

Release	Modification
10.12	Support for the mac parameter introduced for the 4100i Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	<code>config</code> The policy command takes you into the config - policy context where you enter the policy entries.	Administrators or local user group members with execution rights for this command.

policy copy

Syntax

```
policy <POLICY-NAME> copy <DESTINATION-POLICY>
```

Description

Copies a policy to a new destination policy or overwrites an existing policy. Copying a policy copies all its entries as well.

Parameter	Description
<code><POLICY-NAME></code>	Specifies the policy to be copied.
<code><DESTINATION-POLICY></code>	Specifies the name of the destination policy.

Examples

Copying a policy:

```
switch(config)# policy MY_POLICY copy MY_POLICY2
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
      MY_POLICY
2
      my_class3 ipv4
              mirror 1
-----
---
      MY_POLICY2
2
      my_class3 ipv4
              mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

policy resequence

Syntax

```
policy <POLICY-NAME> resequence <STARTING-SEQ-NUM>
                               <INCREMENT>
```

Description

Resequences numbering in a policy.

Parameter	Description
<POLICY-NAME>	Specifies the policy where you want to resequence policy entries.
<STARTING-SEQ-NUM>	Specifies the sequence number to start resequencing from.
<INCREMENT>	Specifies how much to increment the sequence numbers by.

Examples

Resequencing a policy:

```
switch(config)# policy MY_POLICY resequence 1 1
switch(config)# do show policy
      Name
Sequence Comment
      Class Type
              action
-----
---
      MY_POLICY
1
      MY_CLASS3 ipv4
              drop
              dscp AF21
2
      MY_CLASS3 ipv4
              mirror 1
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

policy reset

Syntax

```
policy <POLICY-NAME> reset
```

Description

Changes the user-specified policy configuration to match the active policy configuration. Use this command when a discrepancy exists between what the user configured and what is active and accepted by the system.

Parameter	Description
<POLICY-NAME>	Specifies the policy to be reset.

Examples

Resetting a policy:

```
switch(config)# policy MY_POLICY reset
```

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	config	Administrators or local user group members with execution rights for this command.

show class

Syntax

```
show class [ip | ipv6 | mac] [<CLASS-NAME>] [commands] [configuration]
```

Description

Shows class configuration information.

All parameters are optional.

Parameter	Description
[ip ipv6]	Selects the class type for the display: ip for IPv4, ipv6 for IPv6, or mac for MAC classes.
<CLASS-NAME>	Specifies the class name.
commands	Specifies whether to display output as the CLI commands showing the configured class entries.
configuration	Specifies whether to display classes that have been configured by the user, even if they are not active due to issues with the command parameters or hardware issues. This parameter is useful

Parameter**Description**

l during a mismatch between the entered configuration and the previous successfully programmed (active) classes.

Examples

Showing all class configuration:

```
switch# show class
```

User Configured ipv4 classes:

=====

Type Name

Sequence Comment

action

L3 Protocol

Source IP address

Source L4 Port(s)

Destination IP address or FQDN

Destination L4 Port(s)

Additional Parameters

ipv4 MY_IPV4_CLASS

10 my first class entry comment

match

icmp

192.168.0.1/255.255.255.0

192.168.1.1/255.255.255.0

VLAN: 1

20 my second class entry comment

ignore

tcp

10.100.0.10/255.255.255.0 < 3000

10.100.1.10/255.255.255.0 > 2000

VLAN: 1

Showing class configuration for the IPv4 class MY_IPV4_CLASS as CLI commands:

```
switch# show class ip MY_IPV4_CLASS commands
```

```
class ip "MY_IPV4_CLASS"
```

```
10 match icmp 192.168.0.1/255.255.255.0 192.168.1.1/255.255.255.0 vlan 1
```

```
10 comment my first class entry comment
```

```
20 ignore tcp 10.100.0.10/255.255.255.0 lt 3000 10.100.1.10/255.255.255.0  
gt 2000 vlan 1
```

```
20 comment my second class entry comment
```

Showing class configuration for my_ip_class

```
switch# show class ip my_ip_class
```

Type Name

Sequence Comment

	Action	L3 Protocol
	Source IP Address	Source L4 Port(s)
	Destination IP Address or FQDN	Destination L4
Port(s)	Additional Parameters	

IPv4	my_ip_class	
1	match	icmp
	any	
	www.example.com	
switch# show class ip my_ip_class commands		
class ip my_ip_class		
	1 match icmp any fqdn www.example.com	

Command History

Release	Modification
10.17.1000	Added support for FQDN.
10.12	Support for the mac parameter introduced for the 4100i Switch series.
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

show policy

Syntax

Syntax that shows information for all policies:

```
show policy [commands] [configuration]
```

Syntax that filters by policies applied to an interface or VLAN:

```
show policy [interface <IF-NAME> [in] | vlan <VLAN-ID> [in]]  
[commands] [configuration]
```

Syntax that filters by the named policy:

```
show policy <POLICY-NAME> [commands] [configuration]
```

Syntax that filters by the globally applied policy:

```
show policy global [commands] [configuration]
```

Syntax that shows statistical information in the form of hit counts:

```
show policy hitcounts <POLICY-NAME> [interface <IF-NAME> [in] |  
vlan <VLAN-ID> [in]]
```

Syntax that shows statistical information in the form of hit counts for the globally applied policy:

```
show policy hitcounts global
```

Description

Shows information about your defined policies and where they have been applied. When **show policy** is entered without parameters, information for all policies is shown. The parameters filter the list of policies for which information is shown.

Available filtering includes:

- The content of a specific policy.
- All policies applied to a specific interface.
- All policies applied to a specific VLAN.
- All policies applied to a specific VNI.
- The globally applied policy.

To display policy statistics, use the **show policy hitcounts** form of this command.



NOTE

When a policy is applied to a physical interface or lag using command **apply policy**, with the **per-interface** parameter included, unique instances of the policy are applied to each physical interface port or LAG. The unique instance of a policy has a parent-child relationship with the policy from which it was created. The **show policy** command shows information about the parent policy not the unique instances.



NOTE

If a policy contains any class entries with the count keyword and policy entries with the **cir** action, and the policy is applied to multiple physical or virtual interfaces in the same direction, the statistics will be aggregated. If separate statistics for different physical or virtual interfaces are required, then another policy should be created. Alternatively, in the case of physical interfaces or LAGs, a policy applied with **per-interface** set can be used.

Parameter	Description
<code>interface <IF-NAME></code>	Specifies the interface name.
<code>vlan <VLAN-ID></code>	Specifies the VLAN.
<code>vni<VNI-ID></code>	Specifies the VNI.
<code>in</code>	Selects the inbound (ingress) traffic direction.
<code><POLICY-NAME></code>	Specifies the policy name.
<code>commands</code>	Causes the policy definition to be shown as the commands and parameters used to create it rather than in tabular form.
<code>configuration</code>	Causes the user - configured policies be shown as entered, even if the policies are not active due to policy - definition command issues or hardware issues. This parameter is useful if there is a mismatch between the entered configuration and the previously successfully programmed (active) policies configuration.
<code>global</code>	Selects the globally applied policy.

Parameter	Description
hitcounts	Selects the policy hit counts (statistics).

Examples

Showing information for all policies:

```
switch# show policy
      Name
Sequence Comment
      Class Type
          action
-----
---
      my_policy
10 QOS class
      class1 ipv4
          dscp af21
          drop
20 PBR policy.
      class2 ipv4
          pbr mypbr
-----
---
```

Showing a policy as commands:

```
switch# show policy commands
policy my_policy
      10 class ip class1 action dscp af21 action drop
      20 class ip class2 action pbr mypbr
```

Showing the globally applied policy:

```
switch# show policy global commands
policy global1
      10 class ip my_class1 action drop
apply policy my_policy in
```

Showing policy hit counts (statistics) for the globally applied policy:

```
switch# show policy hitcounts global
Statistics for Policy My_Policy:
global (in):
      Matched Packets  Configuration
10 class ip My_ip_Class
          0  10 match tcp any any ack count
          -  20 match udp any lt 8 any
```

```

        0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
        - 10 match tcp any any ack
        0 20 match icmpv6 1000::10 any count

switch# show policy hitcounts global
Statistics for Policy My_Policy:
global (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

```

Showing policy hit counts (statistics) for a policy applied everywhere (with 1/1/4 and 1/1/5 being applied per interface):

```

switch# show policy hitcounts My_Policy
Statistics for Policy My_Policy:
Interface 1/1/1,lag1 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count
Interface 1/1/4 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count
Interface 1/1/5 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class

```

```

        0 10 match tcp any any ack count
        - 20 match udp any lt 8 any
        0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
        - 10 match tcp any any ack
        0 20 match icmpv6 1000::10 any count
interface 1/1/2.10,1/1/3.10 (in):
    Matched Packets Configuration
10 class ip My_ip_Class
        0 10 match tcp any any ack count
        - 20 match udp any lt 8 any
        0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
        - 10 match tcp any any ack
        0 20 match icmpv6 1000::10 any count
...

```

Showing policy hit counts (statistics) for a policy applied on physical interfaces and LAGs:

```

switch# show policy hitcounts My_Policy interface 1/1/1
Statistics for Policy My_Policy:
Interface 1/1/1,lag1 (in):
    Matched Packets Configuration
10 class ip My_ip_Class
        0 10 match tcp any any ack count
        - 20 match udp any lt 8 any
        0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
        - 10 match tcp any any ack
        0 20 match icmpv6 1000::10 any count

switch# show policy hitcounts My_Policy interface 1/1/1
Statistics for Policy My_Policy:
Interface 1/1/1,lag1 (in):
    Matched Packets Configuration
10 class ip My_ip_Class
        0 10 match tcp any any ack count
        - 20 match udp any lt 8 any
        0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
        0 10 match tcp any any count [ 0 kbps conform ]
        0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

```

Showing policy hit counts (statistics) for a policy applied on VLANs:

```

switch# show policy hitcounts My_Policy vlan 10
Statistics for Policy My_Policy:
vlan 10,20-30 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count
Statistics for Policy My_Policy:
vlan 10,20-30 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count

switch# show policy hitcounts My_Policy vlan 30
Statistics for Policy My_Policy:
vlan 30 (in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

switch# show policy hitcounts My_Policy interface vlan10
Statistics for Policy My_Policy:
VRF red
interface vlan 10,30 (routed-in):
    Matched Packets  Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count

```

```

20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count
show policy hitcounts My_Policy vni 1000
Statistics for Policy My_Policy:
vni 1000 (routed-in):
    Matched Packets Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps
conform ]show policy hitcounts My_Policy vni 1000
Statistics for Policy My_Policy:
vni 1000 (routed-in):
    Matched Packets Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

switch# show policy hitcounts My_Policy interface vlan10
Statistics for Policy My_Policy:
VRF red
interface vlan 10 (routed-in):
    Matched Packets Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

switch# show policy hitcounts My_Policy vrf green routed-in
Statistics for Policy My_Policy:
VRF green
interface vlan 20,25 (routed-in):
    Matched Packets Configuration

```

```

10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed
drop [ 0 kbps conform ]
    - 10 match tcp any any ack
    0 20 match icmpv6 1000::10 any count

switch# show policy hitcounts My_Policy vrf red routed-in
Statistics for Policy My_Policy:
VRF red
interface vlan 10 (routed-in):
    Matched Packets Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]
Statistics for Policy My_Policy:
VRF red
interface vlan 10 (routed-in):
    Matched Packets Configuration
10 class ip My_ip_Class
    0 10 match tcp any any ack count
    - 20 match udp any lt 8 any
    0 30 match icmp any 10.1.1.10 count
20 class ipv6 My_ipv6_Class action cir kbps 1000000 cbs 1000000 exceed drop
    0 10 match tcp any any count [ 0 kbps conform ]
    0 20 match icmpv6 1000::10 any count [ 0 kbps conform ]

```

Command History

Release	Modification
10.08	Added [per - interface] information. Updated examples.
10.07 or earlier	- -

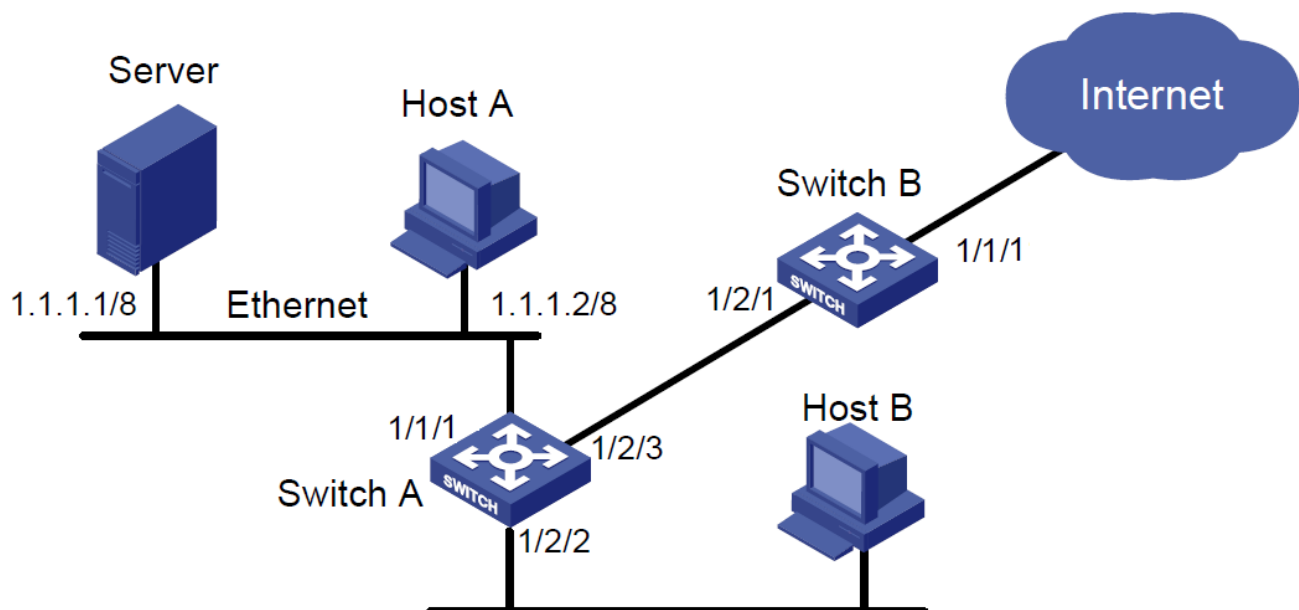
Command Information

Platforms	Command context	Authority
All platforms	Operator (>) or Manager (#)	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

Classifier policies configuration example

This example configures traffic policing on:

- A 10-Gbit Ethernet of Switch A meeting the following requirements:
 - Police the rate of packets from the server to 102,400 kbps. Traffic 102,400 kbps or less is forwarded. The traffic more than 102,400 kbps is dropped.
 - Police the rate of packets from Host A to 25,600 kbps. Traffic 25,600 kbps or less is forwarded. The traffic more than 25,600 kbps is dropped.
- A 10-Gbit Ethernet 1/2/1 of Switch B limiting the incoming traffic rate of HTTP packets on 10-Gbit Ethernet 1/1/1 to the data rate of 204,800 kbps and dropping excess packets.



Subtopics

Configuring the classifier policies example

Configuring the classifier policies example

These steps are part of the classifier policies configuration example.

Procedure

1. Configure Switch A.

Create traffic classes named SERVER_TRAFFIC and HOST_A_TRAFFIC for matching the packets from the server and Host A:

```
switch# configure
switch(config)# class ip SERVER_TRAFFIC
switch(config-class-ip)# match any 1.1.1.1 any
switch(config-class-ip)# exit
switch(config)# class ip HOST_A_TRAFFIC
switch(config-class-ip)# match any 1.1.1.2 any
switch(config-class-ip)# exit
```

2. Create a classifier policy named RATE_LIMIT_POLICY:

```
switch(config)# policy RATE_LIMIT_POLICY
```

3. Configure the policy RATE_LIMIT_POLICY, so that 102,400 kbps of traffic, matching the class SERVER_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy)# class ip SERVER_TRAFFIC action cir kbps 102400
exceed drop
```

4. Configure the policy RATE_LIMIT_POLICY so that 25,600 kbps of traffic, matching the class HOST_A_TRAFFIC, is forwarded and the excess is dropped:

```
switch(config-policy)# class ip HOST_A_TRAFFIC action cir kbps 25600
exceed drop
switch(config-policy)# exit
```

5. Apply RATE_LIMIT_POLICY to interface 1/1/1 for the inbound traffic:

```
switch(config)# int 1/1/1
switch(config-if)# apply policy RATE_LIMIT_POLICY in
switch(config-if)# exit
```

6. To view the configuration with the RATE_LIMIT_POLICY applied:

```

switch# show running-config
Current configuration:
!
...
class ip SERVER_TRAFFIC
    10 match any 1.1.1.1 any
class ip HOST_A_TRAFFIC
    10 match any 1.1.1.2 any
policy RATE_LIMIT_POLICY
    10 class ip SERVER_TRAFFIC action cir kbps 102400 exceed drop
    20 class ip HOST_A_TRAFFIC action cir kbps 25600 exceed drop
interface 1/1/1
    apply policy RATE_LIMIT_POLICY in

```

7. Configure Switch B.

Create a traffic class named HTTP_TRAFFIC and configure it to match traffic to port 80:

```

switch(config)# class ip HTTP_TRAFFIC
switch(config-class-ip)# match tcp any any eq 80
switch(config-class-ip)# exit

```

8. Create a classifier policy named RATE_LIMIT_HTTP:

```

switch(config)# policy RATE_LIMIT_HTTP

```

9. Configure the policy RATE_LIMIT_HTTP so that 204,800 kbps of traffic, matching the class HTTP_TRAFFIC, is forwarded and the excess is dropped:

```

switch(config-policy)# class ip HTTP_TRAFFIC action cir kbps 204800
exceed drop
switch(config-policy)# exit

```

10. Apply RATE_LIMIT_HTTP to interface 1/1/1 for inbound traffic:

```

switch(config)# int 1/1/1
switch(config-if)# apply policy RATE_LIMIT_HTTP in
switch(config-if)# exit

```

11. Show the running configuration with RATE_LIMIT_HTTP applied:

```

switch# show running-config
Current configuration:
!
...
class ip HTTP_TRAFFIC
    10 match tcp any any eq 80

```

```

policy RATE_LIMIT_HTTP
  10 class ip HTTP_TRAFFIC action cir kbps 204800 exceed drop
interface 1/1/1
  apply policy RATE_LIMIT_HTTP in
switch# show running-config
Current configuration:
!
...
class ip HTTP_TRAFFIC
  10 match tcp any any eq 80
policy RATE_LIMIT_HTTP
  10 class ip HTTP_TRAFFIC action cir kbps 204800 exceed drop
interface 1/1/1
  apply policy RATE_LIMIT_HTTP in

```

ACL and Policy hardware resource considerations

Subtopics

Show Resources

TCAM resource consumption and lookups

Matching precedence order

L4 port ranges

Context group selectors

ACL and Policy hardware resource commands

Show Resources

The **show resources** command allows users to see hardware resource consumption in the switch.

These hardware resources include the following:

- TCAM entries
- L4 Port ranges
- High-Capacity TCAM entries

On the 4100i, 6000, 6100, 8320, 8325, and 8360 switch series, these resources are reported for the entire switch. On the 9300 switch series, resources are reported per pipe. The 9300 switch series has four pipes and the 9300s and 10040 switch series has two pipes.

The usage of high capacity TCAM can be viewed specifically using **show system high-capacity-tcam**. This command displays the feature that is currently configured, pending configuration, and the default feature present in high capacity TCAM.

Subtopics

Event Logs

Limitations and exclusions

Event Logs

The following event logs are displayed when the TCAM runs out of resources:

Daemon	Event ID	Severity	Message
ops - switchd	10214	ERROR	TCAM Context Group selectors have been exhausted
ops - switchd	10215	ERROR	TCAM Context Group IDs have been exhausted
ops - switchd	10216	ERROR	Policer resources have been exhausted
ops - switchd	10217	ERROR	TCAM entries have been exhausted
ops - switchd	10218	ERROR	TCAM tables have been exhausted
ops - switchd	10219	ERROR	TCAM ranges have been exhausted
ops - switchd	10220	ERROR	TCAM counters have been exhausted

Limitations and exclusions

On the 4100i, 6000, and 6100 switch series, the widths for **show resources** can have features combined (IPv4 + IPv6) into one lookup. For example: "Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs = 2 TCAM Entry + 4 TCAM Entries = 6 TCAM Entries

Therefore, the table widths for each ACL/Class type are variable depending on what is applied. Widths for each feature: MAC ACL = 1 IPv4 ACL = 2 IPv6 ACL = 4 MAC Class = 1 IPv4 Class = 2 IPv6 Class = 4.

On the 4100i, 6000, 6100, 6300, and 6400 switch series, a MAC Class with an ethertype of 'any' has a width of 7 because it uses one TCAM entry each of MAC, IPv4, and IPv6. Specifying the IPv4 (0x0800) or IPv6 (0x86DD) ethertypes in a MAC Class uses an entry equal to their respective size. IPv4 uses a width of 2 and IPv6 uses a width of 4.

On the 4100i, 6000, 6100, 8320, 8325, and 8400X switch series, resources utilized by the CoPP Lookup are created by default when the system is booted. These entries show up in the used column and may not directly correspond to configured entries.

Resources reserved for one lookup cannot be shared or used by other lookups.

TCAM resource consumption and lookups

TCAM entries and lookups are consumed by features that deal with the classification of packets using the TCAM. TCAM lookups are a finite hardware resource used in the application of ACLs and policies (including port access policies) to packets processed in switch hardware. ADC (analytics data collection) also consumes TCAM lookups. A switch can support a limited number of ACL and policy features at the same time. Use the command **show resources** to monitor TCAM resources and lookups.

Certain platforms have additional resources that features may consume, such as L4 port range checkers, context group selectors, policers, and counters. Different platforms consume these resources at different rates. There are a limited number of features that use TCAM that can be enabled simultaneously on the same line card.



NOTE

In the following TCAM lookup lists, **IP** means both IPv4 and IPv6.

TCAM lookup behaviors vary by switch type.

Subtopics

For the 6000, 6100, and 4100i switch series:

For the 6000, 6100, and 4100i switch series:

The TCAM is uniquely structured on these platforms in a way such that there is one pool of TCAM entries which any feature may consume. Ingress IP CPURX is the only feature that reserves any entries that may only be used by that feature. All other features simply increment their total **Used** TCAM entries by the requested amount. This decrements the total **Free** Ingress/Egress TCAM entries. This is accomplished by labeling each TCAM entry with a lookup ID that corresponds to the feature that entry belongs to.

There are 32 TCAM lookups available to use for these features. Each of these features uses one TCAM lookup when enabled.

- Ingress Global Policy
- Ingress IP Analytics Data Collection (ADC)
- Ingress Port Access Client Policy
- Ingress Port IPv4 ACL
- Ingress Port IPv6 ACL
- Ingress Port MAC ACL
- Ingress Port Policy
- Ingress VLAN IPv4 ACL
- Ingress VLAN IPv6 ACL
- Ingress VLAN MAC ACL
- Ingress VLAN Policy

In this example, IPv6 VLAN ACL, a MAC Port ACL, and a Port Policy with an IPv4 class are configured. IP CPURX entries are installed by default.

```
access-list ip test
  10 deny any any any
  20 deny any any any
  30 deny any any any
access-list ipv6 v6_acl
  10 deny any any any
  20 deny any any any
class ip c
  10 match tcp any any count
policy p
  10 class ip c action drop
vlan 1
vlan 100
  apply access-list ipv6 v6_acl in
interface 1/1/1
  apply access-list mac m in
interface 1/1/2
  apply policy p in
```

[Figure 26](#) illustrates that the Ingress TCAM is one large pool of entries that any feature may consume. The first table in [Figure 26](#) is a **Feature to Lookup ID table** that is used to understand the **Ingress table**. It is not necessary to reserve a chunk of contiguous TCAM entries for a single feature's use.

Figure 1. TCAM Ingress

IP CPURX	0
MAC Port ACL	1
IPv6 VLAN ACL	2
IP Port Policy	3
Ingress TCAM	
0	0
0	0
...	
Reserved by IP CPURX (256)	
0	0
1	3
2	2
2	2
2	2
1	1
1	
...	

```
6100(config)# show resources
Resource Usage:
Mod   Description
Resource                                Used Reserved   Free
-----
---
1/1   Ingress MAC Port ACL Lookup
      Ingress TCAM Entries                8          *
      Ingress IPv6 VLAN ACL Lookup
      Ingress TCAM Entries               12          *
      Ingress IP CPURX Lookup
      Ingress TCAM Entries             158        256
      Ingress Policers                  21
      Ingress IP Port Policy Lookup
      Ingress TCAM Entries                2          *
      Total
```

Ingress TCAM Entries	180	256	3810
Egress TCAM Entries	0	0	2040
Ingress Lookups	4		28
Egress Lookups	0		8
Ingress Policers	21		2027
Egress Policers	0		1024

* This features shares reserved resources with the preceding feature.

Subtopics

Egress TCAM

Egress TCAM

The Egress TCAM operates similarly to the uniquely structured Ingress TCAM on the 4100i, 6000, and 6100 platforms. There is one pool of TCAM entries which any feature may consume.

The following features are configurable on egress:

- Egress Port MAC ACL
- Egress VLAN MAC ACL

Matching precedence order

When a packet is matched by multiple TCAM Lookups with the same action, a precedence order is followed.

For example, if a packet matches an IPv6 Policy with an action to change DSCP to AF11 and a MAC policy with an action to change DSCP to AF12, the MAC DSCP action takes precedence and the DSCP of the packet will change to AF12, given that the precedence of a IPv6 Policy is higher than the precedence of a MAC Policy. Count is an exception in that if a packet matches an IPv4 ACL, MAC ACL, and a policy with count actions, all the counters will increment.

The precedence order from highest to lowest is as follows:

Meter Actions:

Port Access Client Policy

Port Policy

VLAN Policy

Ingress Global Policy

QoS Actions:

Port Access Client Policy Remark

```
Ingress Port Policy Remark
Ingress VLAN Policy Remark
Ingress Global Policy Remark
QoS DSCP Map Entry
QoS COS Map Entry
QoS Port Config
MAC Port ACL Logging
IP Port ACL Logging
MAC VLAN ACL Logging
IP VLAN ACL Logging
```

L4 port ranges

On platforms that support the use of dedicated resources called **L4 Port Ranges**, any ACE that uses **lt**, **gt**, **range**, or port groups attempts to use these dedicated hardware resources. L4 port ranges are used to reduce the number of TCAM entries needed to cover an L4 port range. For example, range 50000 50005 could be covered by a single L4 port range or by two TCAM entries using the following value/masks:

Entry number	L4 port range	Value/Mask
1	range 50000 50003	0xC350/0xFFFC
2	range 50004 50005	0xC354/0xFFFE

The problem gets multiplied when an ACE matches on two L4 port ranges. The TCAM entries used to cover a source and destination L4 port range must cover all possibilities. For example, for this ACE,

```
10 permit tcp any range 100 200 any range 50000 50005
```

The range 100 200 needs six value masks. As a result, this ACE needs 12 (6*2=12) TCAM entries. However, if the TCAM group uses L4 port ranges then the ACE only needs one TCAM entry that matches on two L4 port ranges.

On the 4100i, 6000, 6100, switch series, TCAM entries can share an L4 port range given the following conditions:

- The min and max of the L4 port range are the same
- The L4 port range type (either source or destination L4 ports) is the same
- The TCAM entries are on the same ingress packet processing pipeline

The TCAM entries do not need to be in the same TCAM group.

L4 port ranges are not allocated if the L4 port range can be covered by a single value/mask. For example, **eq 1** and **lt 1024** do not allocate an L4 port range. As a result, the L4 port range allocation can be reduced if

the L4 port range is split into multiple ACEs that can each be covered by a single value/mask. For example, the following ACE:

```
10 permit tcp any any range 50000 50005
```

can be split into these two ACEs with L4 port ranges that can each be covered by a single value/mask:

```
10 permit tcp any any range 50000 50003
```

```
11 permit tcp any any range 50004 50005
```

Reducing the number of L4 port ranges needed by a config can be desirable if the config requires more L4 port ranges than are available in the hardware and there are available TCAM entries to absorb the additional ACEs.

On the 4100i, 6000, and 6100 switch series, there are 60 available L4 port ranges. An L4 port range can be shared between any supported feature.

Context group selectors

There are a limited number of available context group selectors for each context group (Ingress Ports, Ingress VLANs, Egress Ports, Egress VLANs).

Context group selectors work on a first-come-first-served basis. IP ACLs and Classes require two selectors that are allocated together; one selector for each address family (IPv4 and IPv6). Once all the group selectors for a context group have been used, no new application type of ACL or classifier policy for the context group can be applied. For example, if an existing configuration has a MAC ACL, IP ACL, and classifier policy applied on ingress to ports, a policy cannot be applied to a port in the routed-in direction.

Context group selector consumption and availability is as follows:

Type	Selectors
Ingress Port MAC ACL	1
Ingress Port IP ACL	2
Ingress Port Policy	1
Ingress Routed Port Policy	1
Available Ingress Port Selectors	4
-	-
Ingress VLAN MAC ACL	1
Ingress VLAN IP ACL	2

Type	Selectors
Routed - Ingress VLAN IP ACL	2
Ingress VLAN Policy	1
Ingress Routed VLAN Policy	1
Available Ingress VLAN Selectors	4
-	-
Egress Port MAC ACL	1
Egress Port IP ACL	2
Egress Port Policy	1
Available Egress Port Selectors	5
-	-
Egress VLAN MAC ACL	1
Egress VLAN IP ACL	2
Routed - Egress VLAN IP ACL	2
Egress VLAN Policy	1
Available Egress VLAN Selectors	5

ACL and Policy hardware resource commands

Select a command from the list in the left navigation menu.

Subtopics

[show resources](#)

show resources

Syntax

```
show resources
```

Description

Shows hardware resource consumption. Resource data is updated every 10 seconds.

Hardware resource consumption information is shown for:

- TCAM entries
- TCAM lookups
- Policers

Usage

```
"Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entries/  "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entriC(v2) /  "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entries/  "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entriC(v2C(v2)/  "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entries/  "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entriC(v2C(v2) /  "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entries/  "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entriC(v2) /  "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
                        = 5 TCAM entries/  "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
                        = 1 TCAM entry + 4 TCAM entries
```

```

= 5 TCAM entriC(v2C(v2)/ "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entries/ "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entriC(v2C(v2C(v2)/ "Ingress IP Port ACL"
= Ingress v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entries/ "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entriC(v2) / "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entries/ "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entriC(v2C(v2)/ "Ingress IP Port ACL" =
Ingress v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entries/ "Ingress IP Port ACL" = Ingress
v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entriC(v2C(v2C(v2)
MAC ACL 1
IPv4 ACL 1
IPv6 ACL 4
MAC Class 1
IPv4 Class 2
IPv6 Class 4

```

The widths for show resources can have features combined (IPv4 + IPv6) into one TCAM lookup. Therefore, the table widths for each ACL/classifier policy type are variable depending on what is applied. For example:

```

"Ingress IP Port ACL" = Ingress v4 Port ACLs + Ingress v6 Port ACLs
= 1 TCAM entry + 4 TCAM entries
= 5 TCAM entries

```

Widths per feature are as follows:

```

IPv4 ACL 2
MAC ACL 2
IPv6 ACL 4
IPv4 Class 2
IPv6 Class 4

```

Examples

```
switch# show resources
```

```
Resource Usage:
```

```
Mod  Description
```

```
Resource                               Used Reserved    Free
```

```
-----  
1/1  Total
```

```
Ingress TCAM Entries                   0         0  20480
```

```
Egress TCAM Entries                   0         0   8192
```

```
Ingress Lookups                       0          9
```

```
Egress Lookups                       0          4
```

```
Ingress Policers                     0         2048
```

```
Egress Policers                      0         2048
```

```
switch# show resources 1/1
```

```
Resource Usage:
```

```
Mod  Description
```

```
Resource                               Width  Used Reserved    Free
```

```
-----  
1/1  Total
```

```
Ingress TCAM Entries                   0         0  20480
```

```
Egress TCAM Entries                   0         0   8192
```

```
Ingress Lookups                       0          9
```

```
Egress Lookups                       0          4
```

```
Ingress Policers                     0         2048
```

```
Egress Policers                      0         2048
```

```
switch# show resources
```

```
Resource Usage:
```

```
Mod  Description
```

```
Resource                               Used Reserved    Free
```

```
-----  
1/3  Total
```

```
      Ingress TCAM Entries                   0         0  20480
```

```
      Egress TCAM Entries                   0         0
```

```
8192      Ingress Lookups
```

```
0          9
```

```
      Egress Lookups                       0          4
```

```
      Ingress Policers                     0         2048
```

```
      Egress Policers                      0         2048
```

```
1/5  Total
```

```
      Ingress TCAM Entries                   0         0  20480
```

```
      Egress TCAM Entries                   0         0   8192
```

```
      Ingress Lookups                       0          9
```

```
      Egress Lookups                       0          4
```

Ingress Policers	0	2048
Egress Policers	0	2048

switch# **show resources 1/3**

Resource Usage:

Mod	Description	Used	Reserved	Free
Resource				

1/3	Total			
	Ingress TCAM Entries	0	0	20480
	Egress TCAM Entries	0	0	
8192	Ingress Lookups			
0	9			
	Egress Lookups	0		4
	Ingress Policers	0		2048
	Egress Policers	0		2048

switch# **show resources**

Resource Usage:

Mod	Description	Used	Reserved	Free
Resource				

1	Total			
	Ingress TCAM Entries	0	0	18432
	Egress TCAM Entries	0	0	5120
	Ingress Lookups	0		10
	Ingress Flex Loopkups		0	1
	Egress Lookups	0		4
	Ingress Policers	0		5119
	Egress Policers	0		2047

Showing hardware resource consumption on a 6100 switch:

switch# **show resources**

Resource Usage:

Mod	Description	Used	Reserved	Free
Resource				

1/1	Ingress IP CPURX Lookup			
	Ingress TCAM Entries	148	256	
	Ingress Policers	20		
	Total			
	Ingress TCAM Entries	148	256	3832
	Ingress Lookups	1		31
	Ingress Policers	20		2028

Showing hardware resource consumption for line module 1/1 on a 6100 switch:

```
switch# show resources 1/1
```

```
Resource Usage:
```

Mod	Description	Resource	Used	Reserved	Free

1/1	Ingress IP CPURX Lookup				
	Ingress TCAM Entries		148	256	
	Ingress Policers		20		
	Total				
	Ingress TCAM Entries		148	256	3832
	Ingress Lookups		1		31
	Ingress Policers		20		2028

```
switch# show resources
```

```
Resource Usage:
```

Mod	Description	Resource	Width	Used	Reserved	Free

1/1	Ingress IPv4 Port ACL					
	High-Capacity TCAM/LPM Entries		2	0	262144	
	MAC Control Plane Policing					
	TCAM Entries		2	16	256	
	IPv4 Control Plane Policing					
	TCAM Entries		2	70	256	
	IPv6 Control Plane Policing					
	TCAM Entries		2	72	*	
	IPv4 Unicast Route					
	High-Capacity TCAM/LPM Entries		1	0	131072	
	IPv6 Unicast Route					
	High-Capacity TCAM/LPM Entries		2	0	262144	
	IPv4 Multicast Route					
	High-Capacity TCAM/LPM Entries		2	0	65536	
	IPv6 Multicast Route					
	High-Capacity TCAM/LPM Entries		4	0	65536	
	Total					
	TCAM Entries			158	512	
49664						
	High-Capacity TCAM/LPM Entries			0	786432	
258048						
	Policers			0		
65536						
	Ingress L4 Port Ranges			0		24

```
*This feaature shares reserved resources with the preceeding feature.
```

switch# **show resources 1/1**

Resource Usage:

Mod Description

Resource	Width	Used	Reserved	Free

1/1 Ingress IPv4 Port ACL				
High-Capacity TCAM/LPM Entries	2	0	262144	
MAC Control Plane Policing				
TCAM Entries	2	16	256	
IPv4 Control Plane Policing				
TCAM Entries	2	70	256	
IPv6 Control Plane Policing				
TCAM Entries	2	72	*	
IPv4 Unicast Route				
High-Capacity TCAM/LPM Entries	1	0	131072	
IPv6 Unicast Route				
High-Capacity TCAM/LPM Entries	2	0	262144	
IPv4 Multicast Route				
High-Capacity TCAM/LPM Entries	2	0	65536	
IPv6 Multicast Route				
High-Capacity TCAM/LPM Entries	4	0	65536	
Total				
TCAM Entries		158	512	49664
High-Capacity TCAM/LPM Entries		0	786432	258048
Policers		0		65536
Ingress L4 Port Ranges		0		24

* This feature shares reserved resources with the preceding feature.

switch# **show resources**

Resource Usage:

Mod Description

Resource	Width	Used	Reserved	Free

1/1 Global				
Total				
Destination Field Processor Entries			0	0 1024
1/1-0 Ports 9-12,21-24				
Ingress Control Plane Policing				
Ingress TCAM Entries		3	375	6144
Total				
Ingress TCAM Entries			375	6144 18432
VLAN Field Processor Entries			0	0 1024
1/1-1 Ports 1-8				
Ingress Control Plane Policing				

	Ingress TCAM Entries	3	375	6144
Total				
	Ingress TCAM Entries		375	6144
18432				
	VLAN Field Processor Entries		0	0
1024				
1/1-2 Ports 13-20				
	Ingress Control Plane Policing			
	Ingress TCAM Entries	3	375	6144
Total				
	Ingress TCAM Entries		375	6144
18432				
	VLAN Field Processor Entries		0	0
1024				
1/1-3 Ports 25-32				
	Ingress Control Plane Policing			
	Ingress TCAM Entries	3	375	6144
Total				
	Ingress TCAM Entries		375	6144
18432				
	VLAN Field Processor Entries		0	0
1024				

Command History

Release	Modification
10.07 or earlier	- -

Command Information

Platforms	Command context	Authority
4100i	Operator (>) or Manage	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.
6000	r (#)	
6100		

Support and Other Resources

Access HPE Aruba Networking support and updates, and view warranty and regulatory information

Subtopics

[Accessing HPE Aruba Networking Support](#)

[Accessing Updates](#)

[Warranty Information](#)

[Regulatory Information](#)

[Documentation Feedback](#)

Accessing HPE Aruba Networking Support

HPE Aruba Networking Support Services	<u>https://www.hpe.com/us/en/networking/hpe - aruba - networking - support - services.html</u>
AOS - CX Switch Software Documentation Portal	<u>https://arubanetworking.hpe.com/techdocs/ArubaDocPortal/content/new-portal/aoscx.html</u>
HPE Aruba Networking Support Portal	<u>https://networkingsupport.hpe.com/home</u>
North America telephone	1 - 800 - 943 - 4526 (US & Canada Toll - Free Number) +1 - 650 - 750 - 0350 (Backup—Toll Number)
International telephone	<u>https://www.hpe.com/psnow/doc/a50011948enw</u>

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs

- Add-on products or components
- Third-party products or components

Other useful sites

Other websites that can be used to find information:

HPE Aruba Networking Developer Hub	https://developer.arubanetworks.com/hpe - aruba - networking - aoscx/docs/about
Airheads social forums and Knowledge Base	https://community.arubanetworks.com/
AOS - CX Software Technical Update channel on YouTube.	Videos on new features introduced in this release : https://www.youtube.com/playlist?list=PLsYGHuNuBZcbWPEjjHuVMqP - Q_UL3CskS
HPE Aruba Networking Hardware Documentation and Translations Portal	
HPE Aruba Networking software	https://networkingsupport.hpe.com/downloads https://networkingsupport.hpe.com/downloads
Software licensing and Feature Packs	https://licensemanagement.hpe.com/
End - of - Life information	https://networkingsupport.hpe.com/end - of - life

Accessing Updates

You can access updates from the HPE Aruba Networking Support Portal at <https://networkingsupport.hpe.com>.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://networkingsupport.hpe.com/notifications/subscriptions> (requires an active HPE Aruba Networking Support Portal account to manage subscriptions). Security notices are viewable without an HPE Aruba Networking Support Portal account.

Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

Regulatory Information

To view the regulatory information for your product, view the Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

Additional regulatory information

HPE Aruba Networking is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs, product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

Documentation Feedback

HPE Aruba Networking is committed to providing documentation that meets your needs. To help us improve the documentation, send any errors, suggestions, or comments to Documentation Feedback (docsfeedback-switching@hpe.com). When submitting your feedback, include the document title, part number, edition, and publication date located on the front cover of the document. For online help content, include the product name, product version, help edition, and publication date located on the legal notices page.